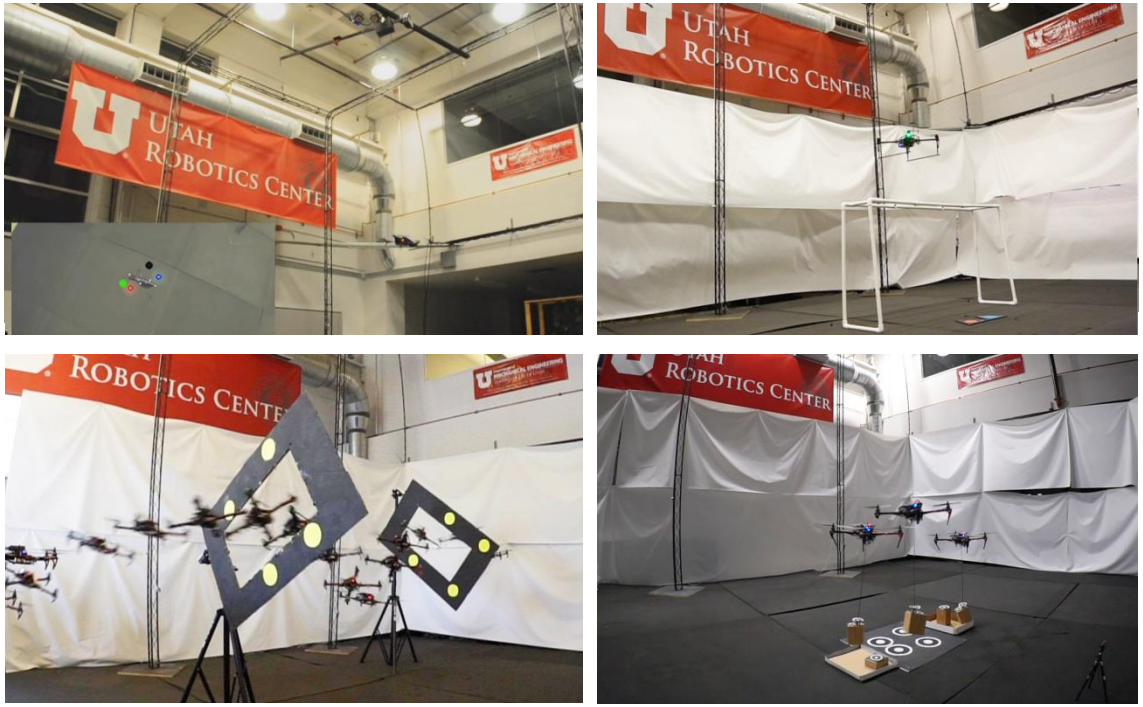
THE UNIVERSITY OF UTAH DARCLab DESIGN, AUTOMATION, ROBOTICS & CONTROL

ROBOTICS CENTER THE UNIVERSITY OF UTAH



Dynamic and Aggressive Image-Based Flying in GPS-Denied Environments: Estimation, Motion Planning, and Control

Dejun Guo
Sr. Engineer

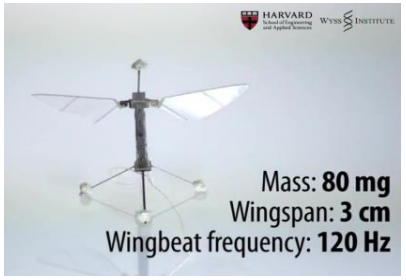
UBTECH Robotics North America R&D Center
Los Angeles, USA

Introduction to aerial robots

Aerial robots and unmanned air vehicles (UAVs):



1. Fixed-wing UAV



2. Flapping-wing design (e.g., Harvard RoboBee)



3. Single-rotor design (e.g., Georgia Tech. autonomous helicopter)



4. Multirotor design (e.g., DJI Phantom 3 quadcopter)

Advantages:

- Readily deployable and low-cost
- Freely moves in 3D space
- Easily carries onboard sensors, cameras, and computational hardware

Applications (Multirotor):



Photography
https://nofilmschool.com/2014/09/faa-decided-allow-studios-to-use-drones-indie-film



Inspection
https://www.inverse.com/article/29388-drone-power-line-china



Precision agriculture
https://factordaily.com/drones-for-precision-agriculture-in-india/



Entertainment
https://dronelife.com/2018/04/20/how-drone-light-shows-are-created-swarm-management-software/



Drone racing
https://www.youtube.com/watch?v=tjF0LdL2Pvk

Motivation and Importance

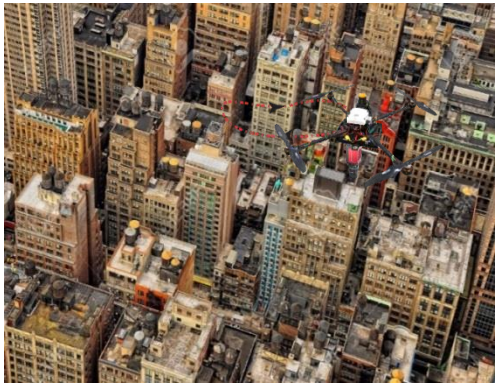
Limitations of aerial robots:

- Autonomous flight and localization relies on an external positioning system (e.g., GPS)
  Goal: use image-based techniques in GPS-denied environments
- Agile motion requires manual control by expert pilots; autopilot restricted to slow maneuvers
  Goal: autonomous aggressive flight control

Examples of GPS-denied environments:



Mountainous terrain



Urban environment



Indoor setting

Importance of proposed work:

- Enables autonomous flight in more complex environments
- Uses cameras as opposed to other on-board localization hardware (e.g., LiDAR)
  - Cameras are cheap and lightweight alternative
  - More information is available in color images than point clouds
- Reduces the need for global sensing modalities for self-localization, motion planning, and control

Applications and challenges

Applications of proposed work:

Search for victims in collapsed buildings





Search and rescue in mountainous areas

Inspect critical infrastructure

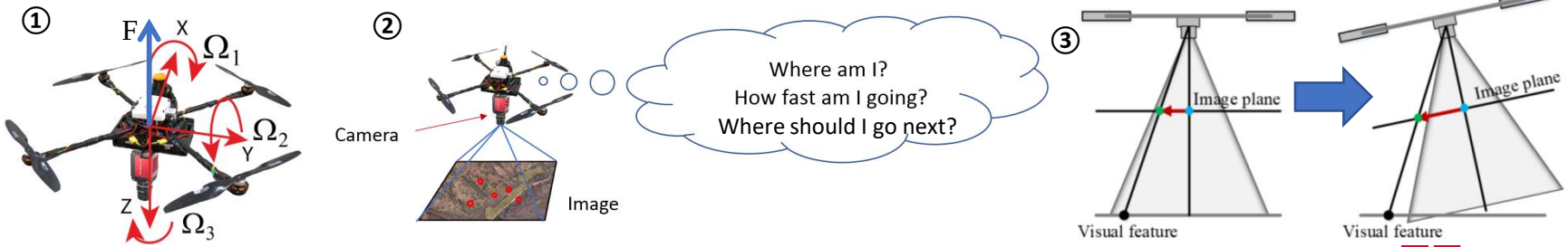




Deliver packages in urban environments

Challenges:

1. Nonlinear dynamics and underactuated mechanisms are inherent in UAV platforms
2. Localization, motion planning, and control with a monocular camera and IMU only is complicated
3. Image measurements are effected by coupled translation and attitude of the robot



State-of-the-art

1. Methods to handle nonlinear dynamics and underactuated mechanisms of UAVs

Nonlinear dynamics

- Linearization -> linear control: PID, LQR, H_∞ — con: small region of attraction
- Nonlinear control
  - Feedback linearization [Lee et. al., 2009] — con: requires accurate plant model
  - Backstepping [Hamel et. al., 2002] — con: limited converging speed
  - Geometry control [Hamel 2002, Mebarki 2015] — con: limited region of attraction

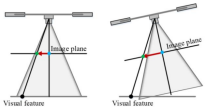
Under-actuated

- Backstepping — pro: rigorous stability analysis — con: limited converging speed
- Hierarchical control loops — pro: fast tracking performance — con: lack rigorous stability analysis
  [Tang et. al., 2018]

2. Sensors for localization, motion planning, and control of UAVs

- GPS [Mahony et. al., 2012] — con: can be blocked
- LiDAR, stereo camera, RGB-D — con: bulky, heavy, expensive
  [Grzonka 2012, Altug 2015, Perez-Grau, 2013]
- Monocular camera — pro: light-weight, low-cost — limitation of prior work: measures position (PBVS)
  [Tang et. al., 2018]

3. Methods to handle the coupling effect of image measurements and robot's attitude

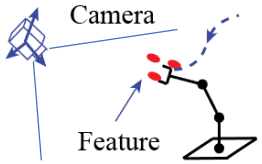


- Gimbal [Thomas et. al., 2016]
- Vision-based pose reconstruction — con: computationally demanding
- Use the data from IMU
  - Virtual image plane [Zheng et. al., 2017] — con: knowledge of target's orientation
  - Unified sphere coordinate [Tahri et. al., 2015] — con: singularity issue

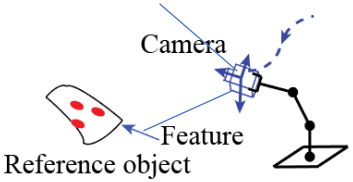
Background of visual servo control

Visual servo control approach:

- **Configuration:**

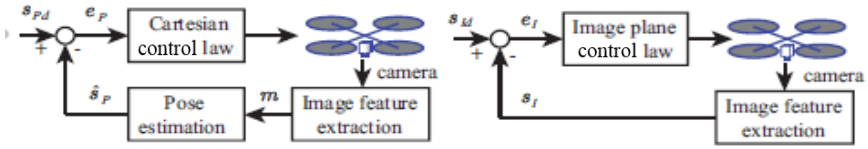


Eye-to-hand
[Liang et. al., 2015]



Eye-in-hand
[Serra et. al., 2016]

- **Structure:**



Position-based visual servo (PBVS) [Falanga et. al., 2017]

Image-based visual servo (IBVS) [Mebarki et. al., 2017]

Pros:
Observe entire motion space

Cons:
Motion space is limited by the view

Pros:
Unlimited motion space

Cons:
- Camera motion is coupled with robot motion
- Camera view is limited

Pros:
Smooth trajectory in space

Cons:
- Need pose estimation
- Sensitive to measurement noise

Pros:
- No pose estimation (Simple structure, lower computational demand)
- Robust to measurement noise

Cons:
- Nonlinear model
- Singularity
- Complex stability analysis

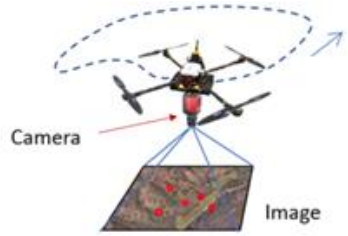
Overview of my work

What I've contributed: (1) developed image-based algorithms for estimation, planning, and control of aerial robots using a single camera; (2) provided rigorous stability and performance analysis; and (3) validated through experiments

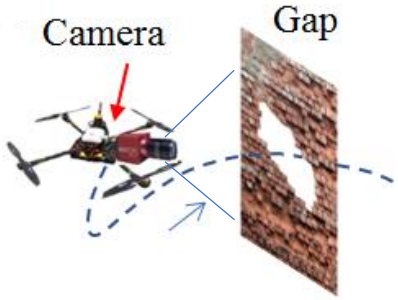


Mobile camera √

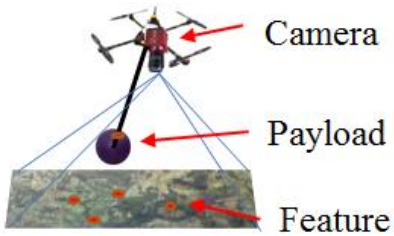
Step1: estimate and control through a mobile overhead camera (Eye-to-hand configuration + IBVS)



Step2: localize and control through an onboard camera (Eye-in-hand configuration + PBVS)



Step3: motion plan and control for aggressive maneuvering through openings (Eye-in-hand configuration + IBVS)



Step4: motion plan and control to carry payloads (Eye-in-hand configuration + IBVS)



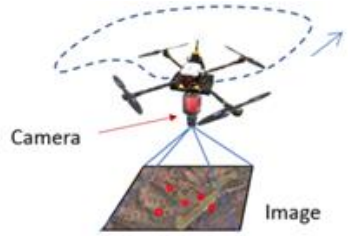
Overview of my work

What I've contributed: (1) developed image-based algorithms for estimation, planning, and control of aerial robots using a single camera; (2) provided rigorous stability and performance analysis; and (3) validated through experiments

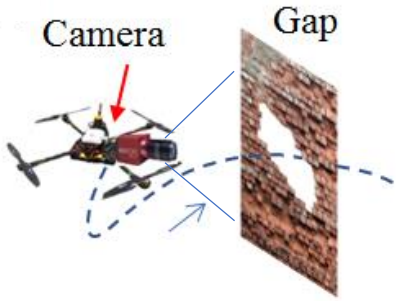


Mobile camera √

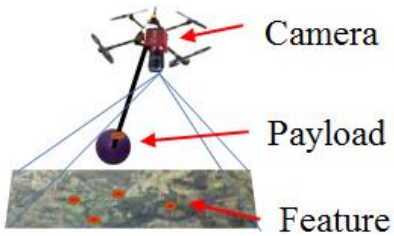
Step1: estimate and control through a mobile overhead camera (Eye-to-hand configuration + IBVS)



Step2: localize and control through an onboard camera (Eye-in-hand configuration + PBVS)



Step3: motion plan and control for aggressive maneuvering through openings (Eye-in-hand configuration + IBVS)

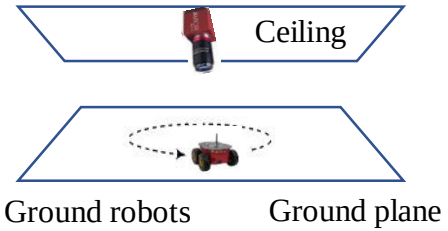


Step4: motion plan and control to carry payloads (Eye-in-hand configuration + IBVS)

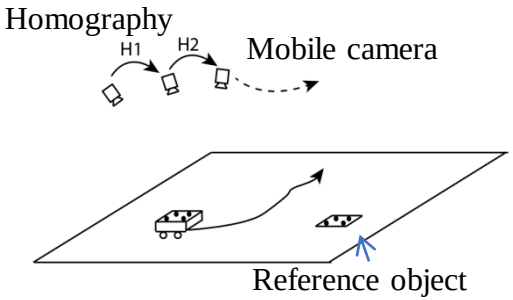


Step 1: estimation and control in eye-to-hand configuration

Prior work and their limitations:



- Stationary overhead camera → Ground robots move in a plane;
  - **Methods**: Adaptive control [Dixon et. al., 2001], etc.
  - **Limitations**: Motion is limited in a plan (to simplify depth and control design (not practical for 3D motion))



- Mobile overhead camera → Ground robots move in a plane;
  - **Methods**: Homography [Lopez-Nicolas et. al., 2011]
  - **Limitations**: ▪ Requires accurate camera calibration
    ▪ Does not reject periodic errors

- Dealing with uncertain system parameters
  - **Methods**: Estimating Jacobian [Liu et. al., 2005]; Adaptive [Braganza et. al., 2008]
  - **Limitations**: Assume invariant or slow-varying depth

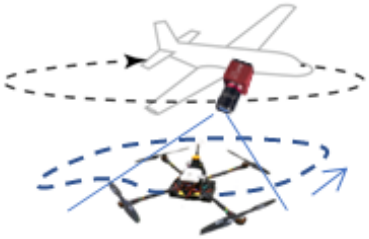
- Handling periodic tracking errors
  - **Methods**: Iterative learning control (ILC) [Hehn et. al., 2014]
  - **Limitations**: Require resetting initial conditions (not practical)
  - **Methods**: Repetitive control (RC) [Ott et. al., 2008]
  - **Limitations:** No stability analysis for the nonlinear visual servo approach

Knowledge gap: No methods can control an aerial robot through the mobile overhead camera and deal with uncertainties and periodic errors simultaneously

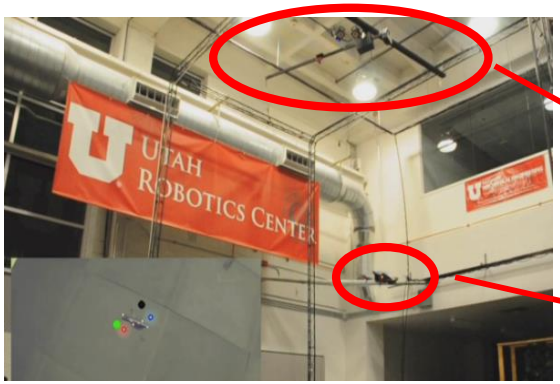
Step 1: estimation and control in eye-to-hand configuration

My approach: Localize and control an aerial robot through a mobile overhead camera by exploiting image-based adaptation and repetition. (Eye-to-hand configuration + IBVS)

My results:



Mobile camera √



Mobile overhead camera on a rotating arm

Aerial robot

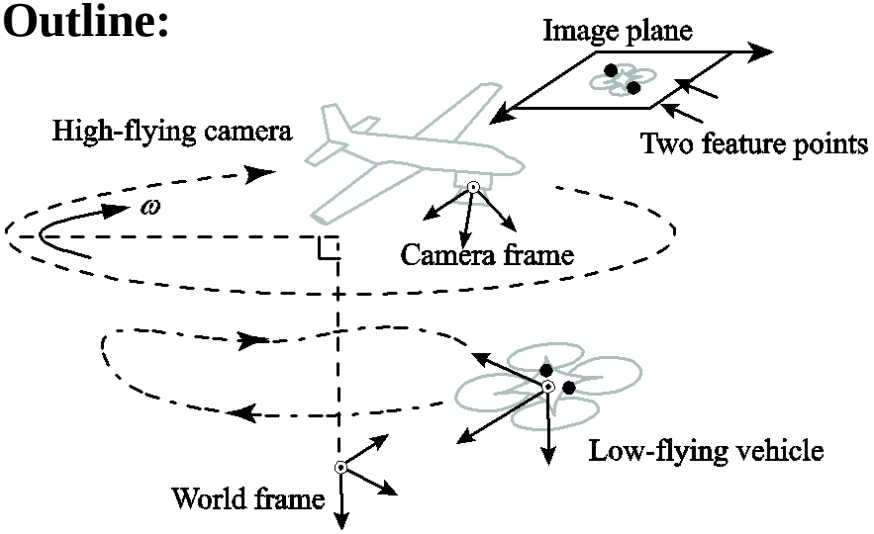
Image from overhead camera

Contribution: First demonstration of motion control in GPS-denied environments using a moving overhead camera to observe and monitor the scene, through the eye-to-hand configuration. Stability analysis is presented.

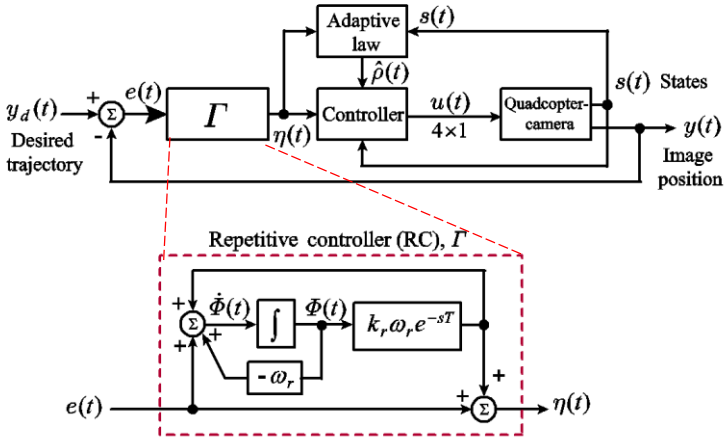
Step 1: estimation and control in eye-to-hand configuration

Localize and control an aerial robot through a mobile overhead camera by exploiting image-based adaptation and repetition.

Outline:



High-flying camera for visual-servo control of a low-flying vehicle



Adaptive repetitive visual-servo controller

- Adaptive estimator handles model uncertainty
- Repetitive controller tracks periodic trajectory

Related publications:

[J1] **D. Guo**, J. Bourne, H. Wang, W. Yim, and K. K. Leang, "Adaptive-repetitive visual-servo control of low-flying aerial robots via uncalibrated high-flying cameras", *Journal of Nonlinear Science, Special Issue on Robotics: Mechanics and Control of Locomotion*, 27(4): 1235 – 1256, 2017.

[C1] **D. Guo**, W. Yim, and K. K. Leang, "Adaptive repetitive visual-servo control of a low-flying unmanned aerial vehicle with an uncalibrated high-flying camera", *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Daejeon, Korea, Oct. 9-13, 2016.

[C2] X. He, **D. Guo**, and K. K. Leang, "Repetitive control design and implementation for periodic motion tracking in aerial vehicle", *American Control Conference (ACC)*, Seattle, WA, USA, May 24-26, 2017.

Step 1: estimation and control in eye-to-hand configuration

Adaptive-repetitive control system design and stability analysis

- **Camera model** (perspective):

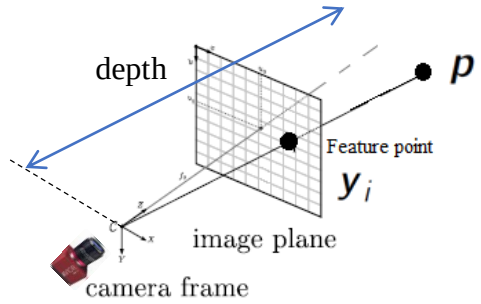
Camera parameters

Feature position in the camera frame

$$\begin{bmatrix} y_i \\ 1 \end{bmatrix} = \frac{1}{z_i}\mathbf{H}_i \begin{bmatrix} p \\ 1 \end{bmatrix}$$

Image position

Depth



- **Adaptive estimator:**

Recall Kinematics:

$$\begin{cases} \dot{y}_i = z_i^{-1}\mathbf{J}_i u + \mathbf{A} \\ \dot{z}_i = \mathbf{F}_i u + \mathbf{B} \end{cases}$$

Linearly parameterized:

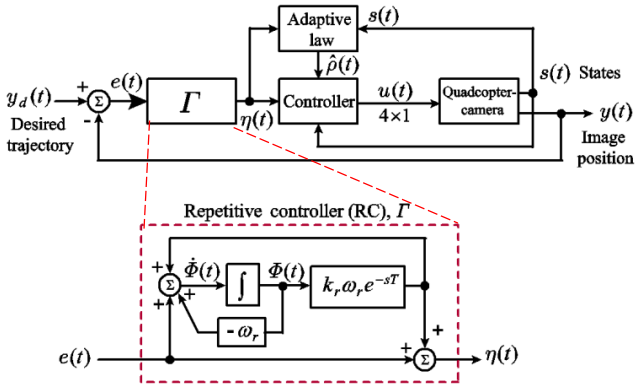
$$\mathbf{J}_i u = \mathbf{C}_i \rho$$
$$\mathbf{F}_i u = \mathbf{Q}_i \rho$$

Estimation errors

Adaptive estimator for

$\hat{\rho}$

Camera parameters



- **Image-based controller:**

$$u = \hat{\mathbf{D}}^{-1}(-\mathbf{K}_p\eta - \hat{\mathbf{Z}}\Omega - k_s\Phi)$$

Related to Jacobian

Control gains

Depths

Assume lower-lever servo control can achieve u immediately

- **Stability analysis:** the system is asymptotically stable

positive-definite storage function:

$$V = 1/2(\eta^T\mathbf{Z}\eta + \Phi^T\Phi + \tilde{\rho}^T\mathbf{\Gamma}\tilde{\rho})$$

Substitute with closed-loop system:

$$\dot{V} \leq 0$$

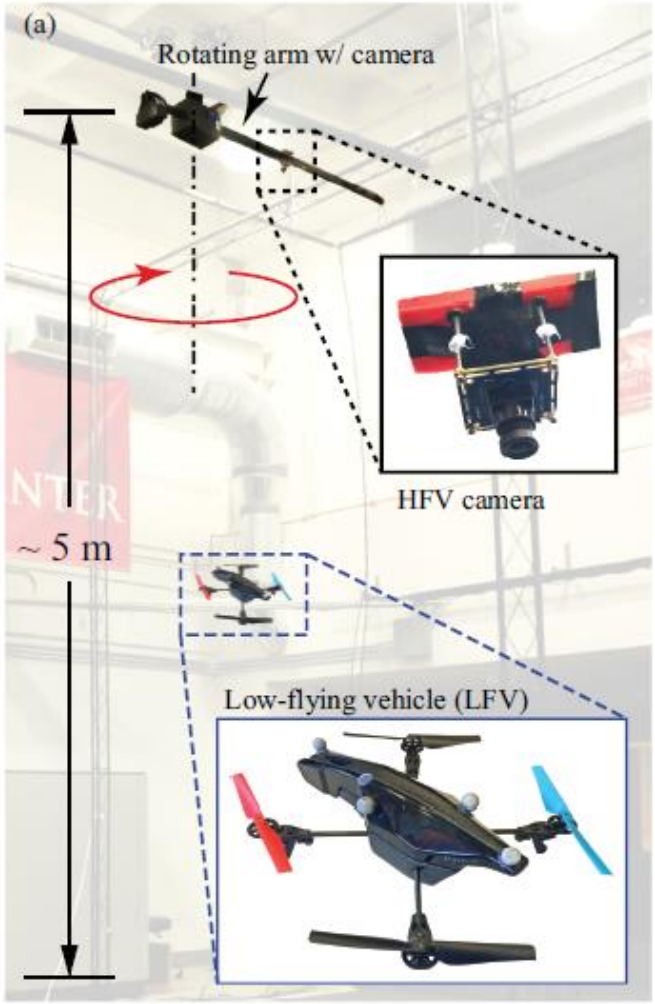
Barbalat's Lemma:

$$\dot{V} \to 0 \text{ as } t \to \infty$$

$$e \to 0$$

Step 1: estimation and control in eye-to-hand configuration

Performance is validated through simulations and physical experiments







Step 1: estimation and control in eye-to-hand configuration

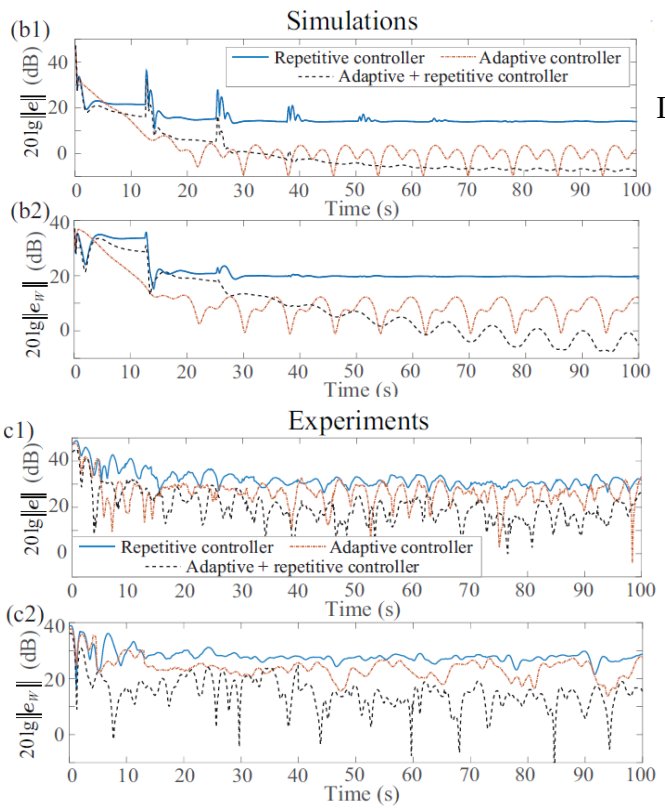
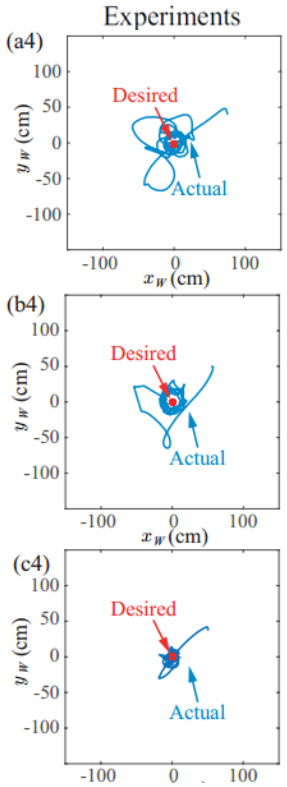


Image errors

RMS errors

Image errors

RMS errors



Repetitive controller

Adaptive controller

Adaptive repetitive controller

Trajectory in the world frame

Case1: Hovering in place

Step 1: estimation and control in eye-to-hand configuration

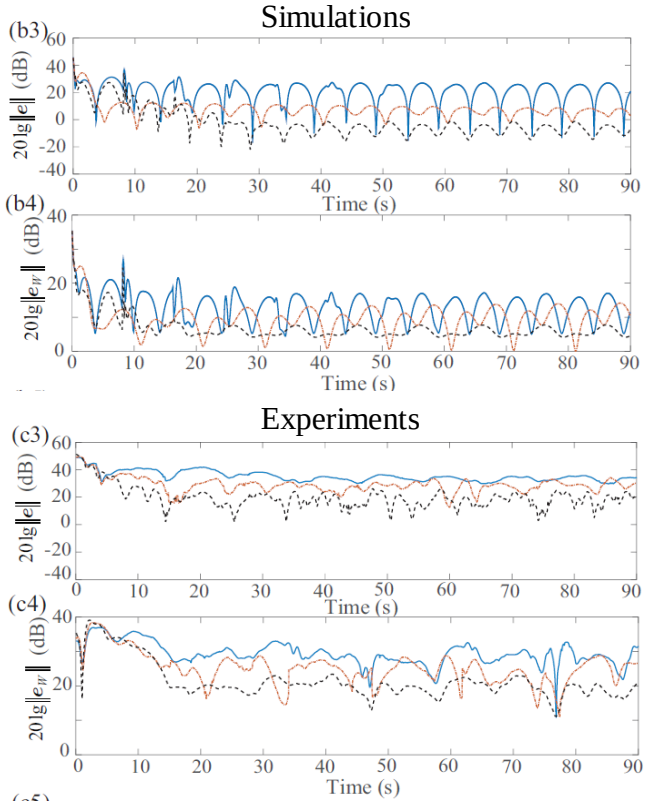
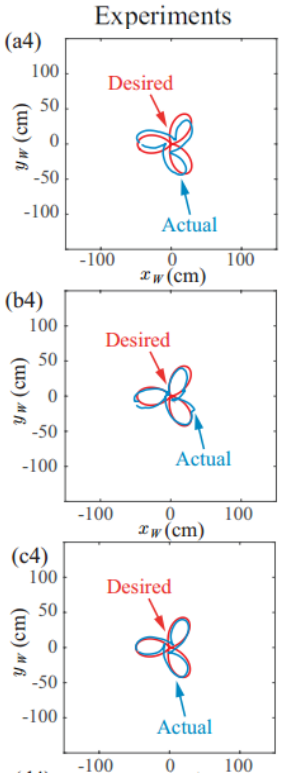


Image errors

RMS errors

Image errors

RMS errors



Repetitive controller

Adaptive controller

Adaptive repetitive controller

Trajectory in the world frame

Case2: Tracking flower trajectory

Step 1: estimation and control in eye-to-hand configuration

(a) Simulation		Image variables $\boldsymbol{y}(t)$		Pose variables	
		e_max(%)	e_rms(%)	e_max(%)	e_rms(%)
Hover (case 1)	Repetitive	0.46	0.44	1.12	1.12
	Adaptive	0.13	0.09	0.47	0.32
	Repetitive+adaptive	0.04	0.04	0.13	0.07
Flower traj. (case 2)	Repetitive	2.04	1.30	0.84	0.56
	Adaptive	0.29	0.19	0.56	0.39
	Repetitive+adaptive	0.08	0.05	0.28	0.22

(b) Experiment		Image variables $\boldsymbol{y}(t)$		Pose variables	
		e_max(%)	e_rms(%)	e_max(%)	e_rms(%)
Hover (case 1)	Repetitive	4.54	3.14	3.56	2.70
	Adaptive	4.00	1.24	2.87	1.04
	Repetitive+adaptive	1.83	0.83	1.05	0.60
Flower traj. (case 2)	Repetitive	5.85	4.17	5.00	3.10
	Adaptive	4.58	1.35	3.22	1.21
	Repetitive+adaptive	1.78	0.81	1.71	1.15

Average RMS tracking error is reduced by approximately **66**% in the image plane and over **71**% in the world frame.

Conclusion: Compared with to standard RC and adaptive, the proposed adaptive-repetitive control has superior performance.

Related publications:

[J1] **D. Guo**, J. Bourne, H. Wang, W. Yim, and K. K. Leang, "Adaptive-repetitive visual-servo control of low-flying aerial robots via uncalibrated high-flying cameras", *Journal of Nonlinear Science, Special Issue on Robotics: Mechanics and Control of Locomotion*, 27(4): 1235 – 1256, 2017.

[C1] **D. Guo**, W. Yim, and K. K. Leang, "Adaptive repetitive visual-servo control of a low-flying unmanned aerial vehicle with an uncalibrated high-flying camera", *IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**)*, Daejeon, Korea, Oct. 9-13, 2016.

[C2] X. He, **D. Guo**, and K. K. Leang, "Repetitive control design and implementation for periodic motion tracking in aerial vehicle", *American Control Conference (**ACC**)*, Seattle, WA, USA, May 24-26, 2017.

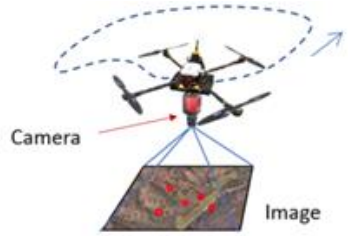
Overview of my work

What I've contributed: (1) developed image-based algorithms for estimation, planning, and control of aerial robots using a single camera; (2) provided rigorous stability and performance analysis; and (3) validated through experiments

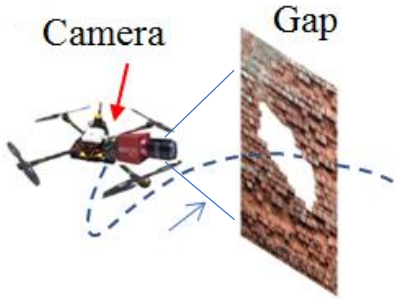


Mobile camera √

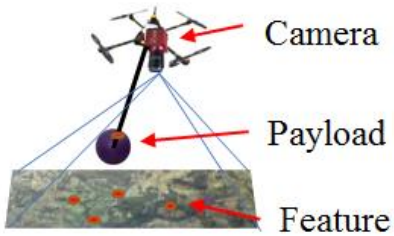
Step1: estimate and control through a mobile overhead camera (Eye-to-hand configuration + IBVS)



Step2: localize and control through an onboard camera (Eye-in-hand configuration + PBVS)



Step3: motion plan and control for aggressive maneuvering through openings (Eye-in-hand configuration + IBVS)



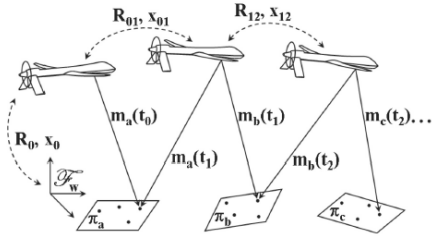
Step4: motion plan and control to carry payloads (Eye-in-hand configuration + IBVS)



Step 2: localization and control in eye-in-hand configuration

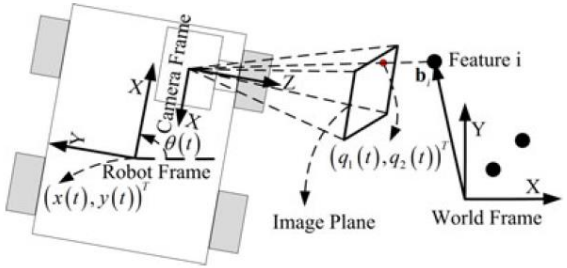
Prior work and their limitations:

- Onboard camera → 3D state estimation
  - **Methods**: Visual SLAM [Achtelik. et. al., 2011], visual-inertial odometry (VIO) [Kaiser, et. al., 2010, Zhao. et. al., 2016].
  - **Limitations**: ▪ Is computationally demanding
    ▪ Requires zero initial estimated errors
    ▪ Assumes the controller and estimation process to be stable
    (dynamic effects of estimation may deteriorate the control performance)



VIO framework
[Kaiser, et. al., 2010]

- Onboard camera → 2D State estimation and control + stability analysis
  - **Methods**: Nonlinear observer [Wang, et. al., 2014, Liang et. al., 2016]
  - **Limitations**: ▪ Kinematics
    ▪ Constrained in a plane
    ▪ Planar motion simplifies depth and control design (not practical for 3D motion)



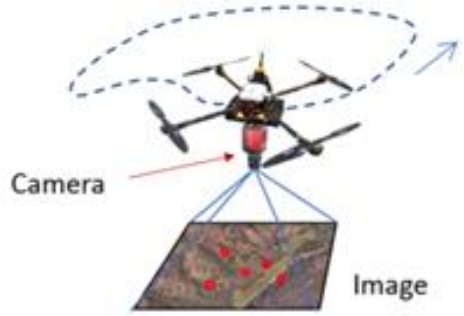
Observer-based estimation
[Wang, et. al., 2014]

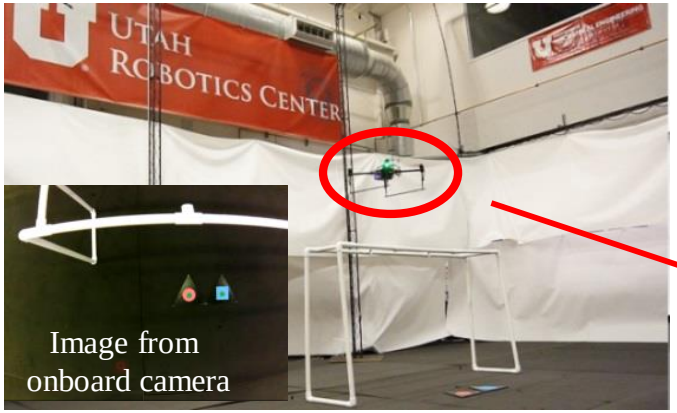
Knowledge gap: The stability analysis for closed-loop observer-controller system for robots moving in 3D space has received limited attention.

Step 2: localization and control in eye-in-hand configuration

My approach: Localize and control an aerial robot using an onboard camera by developing a closed-loop vision-based observer and backstepping controller. (Eye-in-hand configuration + PBVS)

My results:



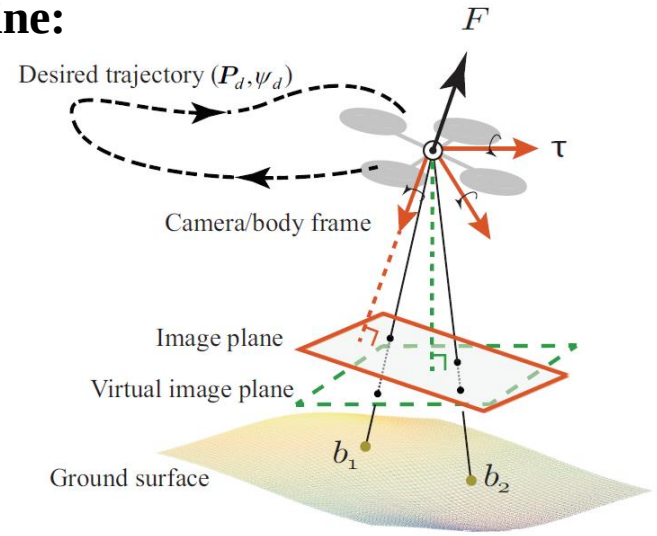


Contribution: First demonstration of localization and motion control with rigorous stability analysis in GPS-denied environments using an onboard camera, through the eye-in-hand configuration.

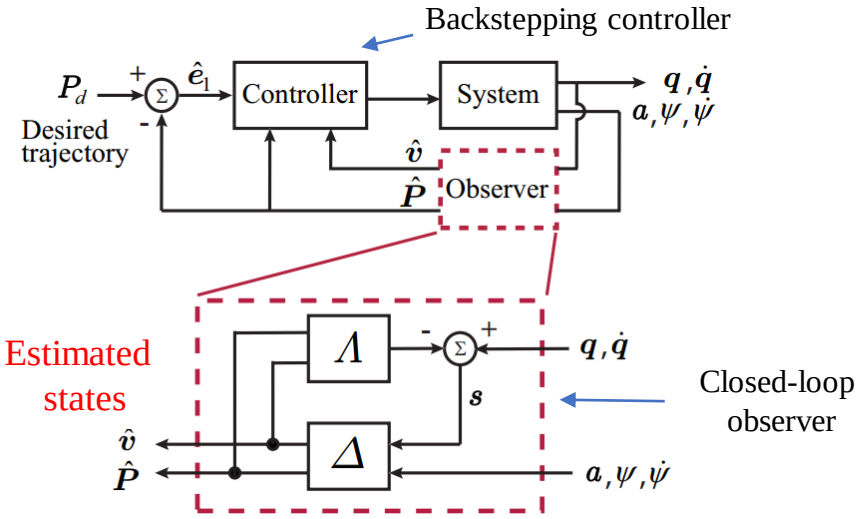
Step 2: localization and control in eye-in-hand configuration

Localize and control an aerial robot using an onboard camera by developing a closed-loop vision-based observer and backstepping controller

Outline:



Position and linear velocity estimation for position-based visual servo control of an aerial robot



Backstepping controller & closed-loop observer

- Closed-loop observer estimates the uncertain positions and linear velocities
- Backstepping controller stabilizes uncontrolled states

Related publications:

[J2] **D. Guo**, H. Wang, and K. K. Leang. "Nonlinear vision-based observer for visual servo control of an aerial robot in GPS-denied environments", *ASME Journal of Mechanisms and Robotics*, 10(6), 061017-061017-13, 2018.

[C3] **D. Guo** and K. K. Leang. "Position and linear velocity estimation for position-based visual servo control of an aerial robot in GPS-denied environments", *ASME Dynamic Systems and Control Conference (DSCC)*, Tysons Corner, Virginia, USA, Oct. 11-13, 2017. (**This paper won the 2017 ASME DSCC Student Best Paper Award**)

Step 2: localization and control in eye-in-hand configuration

Virtual image plane + perspective camera model:
(decouple robot attitude from the image)

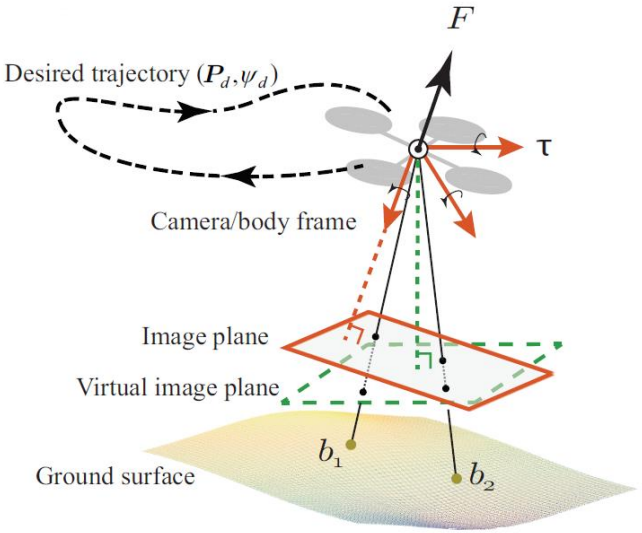
y_i in image plane —IMU→ q_i in virtual image plane

Virtual image position

$$q_i = \frac{1}{d_i}\begin{bmatrix} m_1 \\ m_2 \end{bmatrix} \mathbf{R}_\psi^T (b_i - P),$$

Virtual depth; Intrinsic parameters; Rotation matrix by yaw; Coordinate of feature points in camera frame

$$d_i = m_3 \mathbf{R}_\psi^T (b_i - P),$$



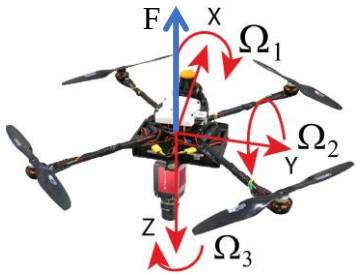
Dynamic model:

Linear velocity in the body frame

Translational $m\dot{\boldsymbol{v}} = -m\boldsymbol{\omega}\times\boldsymbol{v} - f\boldsymbol{E}_3 + mg\mathbf{R}^T\boldsymbol{E}_3$ — Force

Angular $\mathcal{I}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega}\times\mathcal{I}\boldsymbol{\omega} + \boldsymbol{\tau}$ — Torque

Angular velocity



Step 2: localization and control in eye-in-hand configuration

Observer-controller design and stability analysis

- **Closed-loop observer**

Define extended estimated projection error:

$$\begin{cases} s_{1i} = \hat{d}_i q_i - \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} \mathbf{R}_\psi^T (b_i - \hat{P}) \\ s_{2i} = \dot{\hat{d}}_i q_i + \hat{d}_i \dot{q}_i - \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} (-sk(\dot{\psi}) \mathbf{R}_\psi^T (b_i - \hat{P}) - \hat{v}) \end{cases}$$

Linearly parameterized:

$$s_i = \begin{bmatrix} s_{1i} \\ s_{2i} \end{bmatrix} = \underbrace{\begin{bmatrix} G_{i1} \\ G_{i2} \end{bmatrix}}_{G_i \in \Re^{4\times 6}} \begin{bmatrix} \tilde{P} \\ \tilde{v} \end{bmatrix}$$

Estimation errors of position and velocity

If $G = \left[G_1^T, G_2^T\right]^T \in \Re^{8\times 6}$ is full column rank, then $\tilde{P}, \tilde{v} \to 0$, as $s \to 0$.

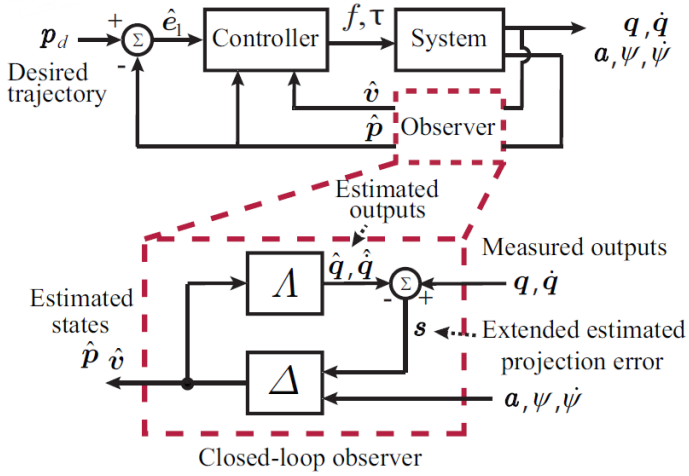
Closed-loop observer:

$$\dot{\hat{P}} = \mathbf{R}_\psi \hat{v} - \mathbf{L}_p^{-1} (\mathbf{G}_{1-3}^T \mathbf{K} s - \mathbf{K}_1 c_1 \hat{e}_1),$$

$$\dot{\hat{v}} = -sk(\dot{\psi}) \hat{v} + a - \mathbf{L}_v^{-1} (\mathbf{G}_{4-6}^T \mathbf{K} s + \mathbf{K}_1 \hat{e}_1),$$

Gains

Guarantee $\hat{P}$ and $\hat{v}$ converge to the truth values



Step 2: localization and control in eye-in-hand configuration

Observer-controller design and stability analysis

- **Backstepping controller:**

Difference between estimated and desired states

$$u\mathbf{R}_{\phi\theta}\boldsymbol{E}_3 = m(g\boldsymbol{E}_3 - c_1\eta - c_1k_2\boldsymbol{e}_2)$$

Thrust — Roll, pitch — Hierarchical errors

Assume lower-lever servo control can achieve u, ϕ, and θ immediately

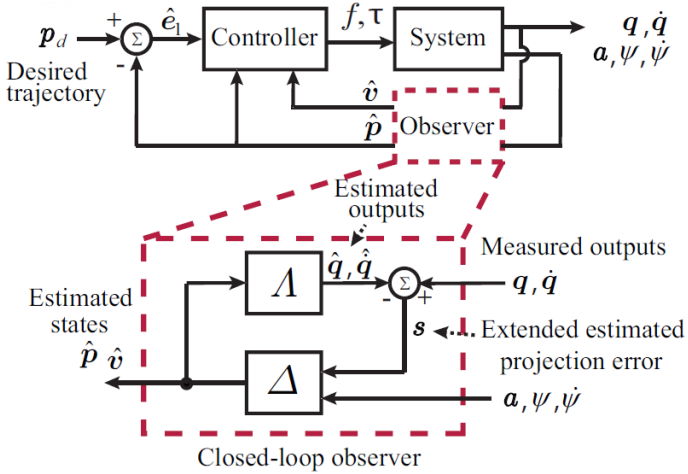
- **Stability analysis:** the system is asymptotically stable

positive-definite storage function:

$$V = 1/2(\boldsymbol{e}_1^T\mathbf{K}_1\boldsymbol{e}_1 + \boldsymbol{e}_2^T\boldsymbol{e}_2 + \boldsymbol{e}_3^T\boldsymbol{e}_3 + e_\psi^2 + \tilde{\boldsymbol{\rho}}^T\mathbf{L}\tilde{\boldsymbol{\rho}})$$

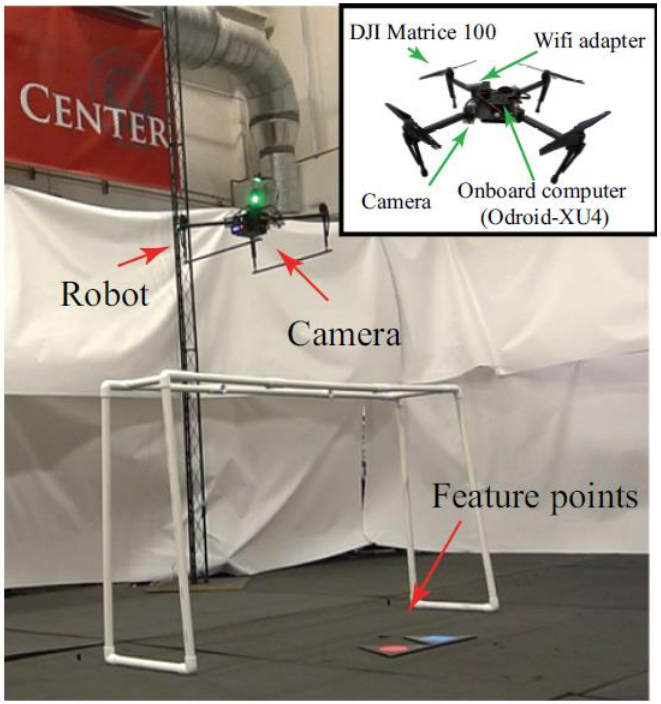
$\dot{V} \le 0$, Barbalat's Lemma: $\dot{V} \to 0$ as $t \to \infty$

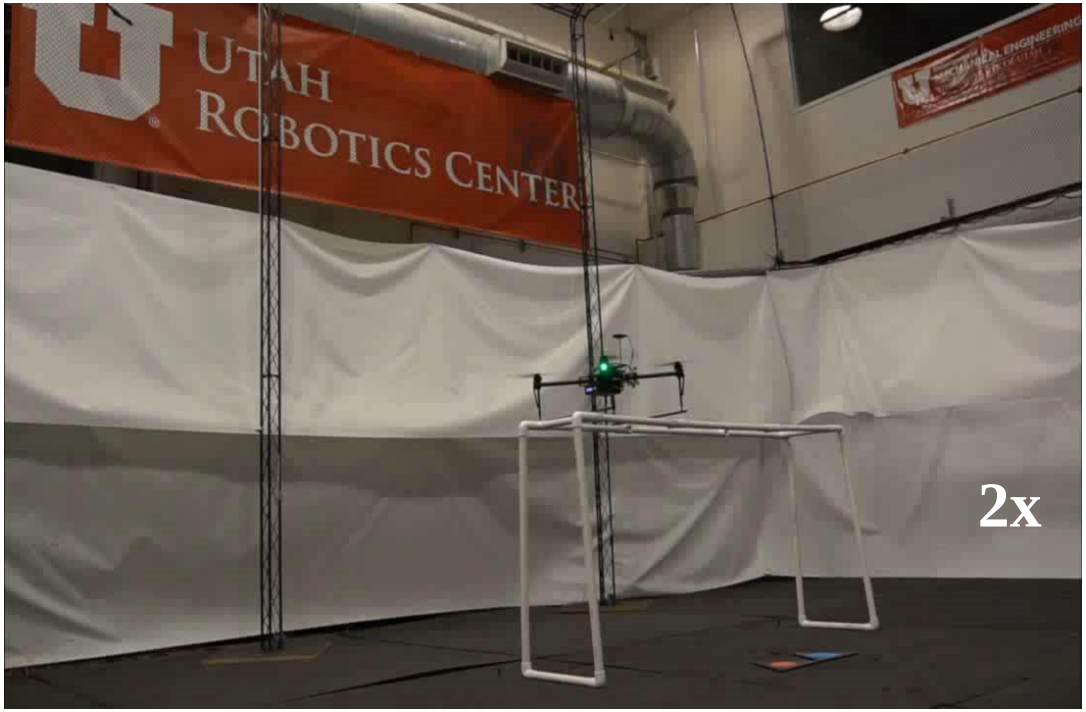
$\tilde{\boldsymbol{\rho}} \to 0 \quad \boldsymbol{e}_1 \to 0$ as $t \to \infty$



Step 2: localization and control in eye-in-hand configuration

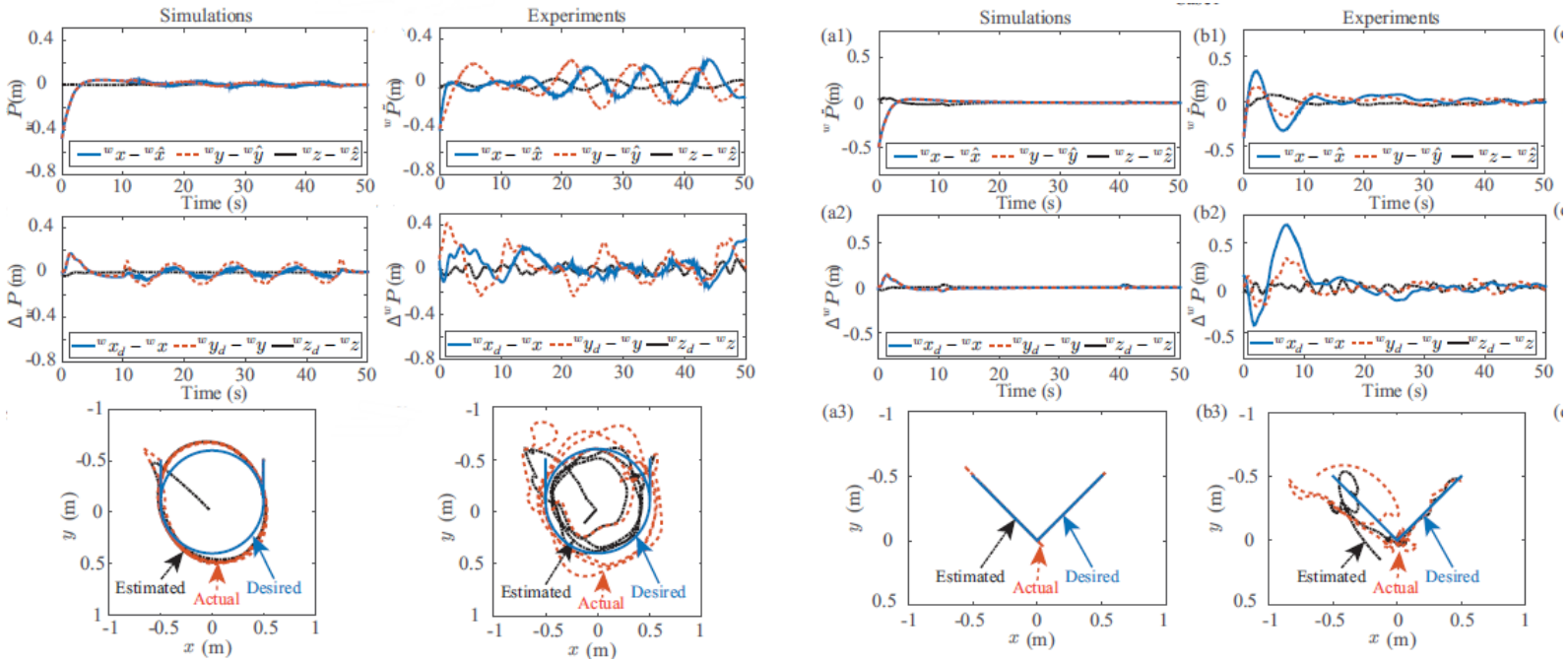
Performance is validated through simulations and physical experiments





Circular trajectory

Step 2: localization and control in eye-in-hand configuration



Conclusions:

1. The proposed closed-loop observer and backstepping controller stabilize the system.
2. The position converges to the desired trajectory and the estimated states converge to the true values.

Related publications:

[J2] **D. Guo**, H. Wang, and K. K. Leang. "Nonlinear vision-based observer for visual servo control of an aerial robot in GPS-denied environments", *ASME Journal of Mechanisms and Robotics*, 10(6), 061017-061017-13, 2018.

[C3] **D. Guo** and K. K. Leang. "Position and linear velocity estimation for position-based visual servo control of an aerial robot in GPS-denied environments", *ASME Dynamic Systems and Control Conference (DSCC)*, Tysons Corner, Virginia, USA, Oct. 11-13, 2017. (**This paper won the 2017 ASME DSCC Student Best Paper Award**)

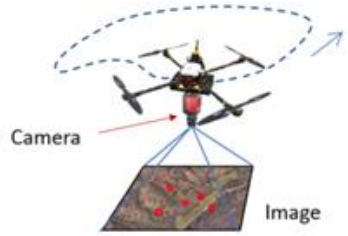
Overview of my work

What I've contributed: (1) developed image-based algorithms for estimation, planning, and control of aerial robots using a single camera; (2) provided rigorous stability and performance analysis; and (3) validated through experiments

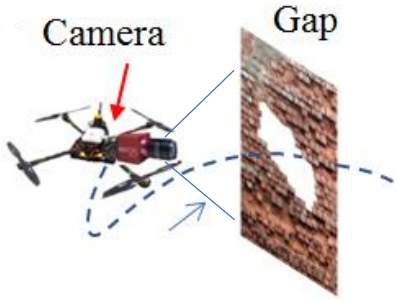


Mobile camera √

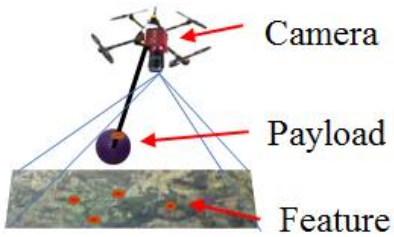
Step1: estimate and control through a mobile overhead camera (Eye-to-hand configuration + IBVS)



Step2: localize and control through an onboard camera (Eye-in-hand configuration + PBVS)



Step3: motion plan and control for aggressive maneuvering through openings (Eye-in-hand configuration + IBVS)



Step4: motion plan and control to carry payloads (Eye-in-hand configuration + IBVS)



Step 3: aggressive maneuvers in eye-in-hand configuration

Prior work and their limitations:

- Onboard camera → Position-based aggressive maneuvers



[Thomas. et. al., 2014]

- **Methods**: Virtual image plane + Image-based geometry control [Thomas. et. al., 2014],
- **Limitations**: ▪ Prior knowledge of reference object's model (orientation and radius)
  → formulate a one-to-one function of image projection → simplify motion planning
  ▪ Constrained in a 2D vertical plane



[Falanga. et. al., 2017]

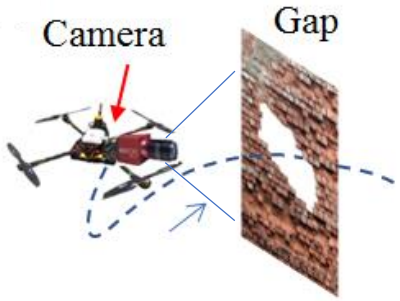
- **Methods**: Pose estimation + Geometry control [Falanga. et. al., 2017],
- **Limitations**: ▪ Prior knowledge of reference object's model (window's size)
  ▪ Sensitive to image noise

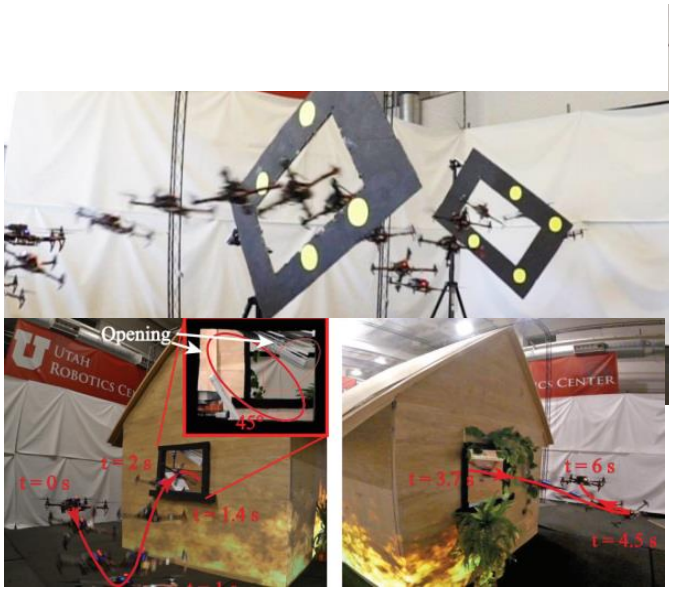
Knowledge gap: Limited approaches enable aggressive maneuvers in 3D space only using an onboard camera and without considering translational variables.

Step 3: aggressive maneuvers in eye-in-hand configuration

My approach: Image-based motion planning and control of aerial robots for traverse through multiple openings. (Eye-in-hand configuration + IBVS)

My results:



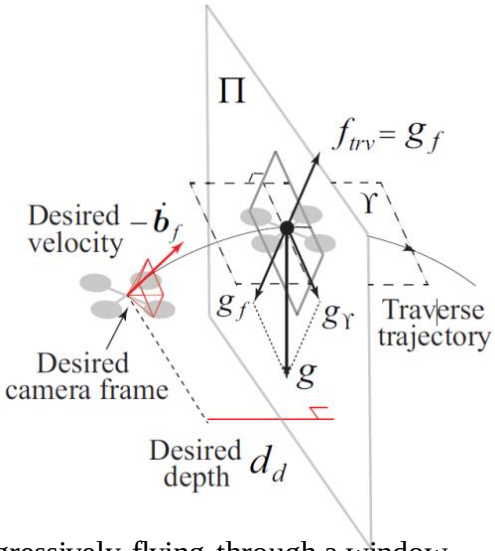


Contribution: First demonstration of real-time motion planning and control in the image coordinate of an onboard camera to fly through narrow openings

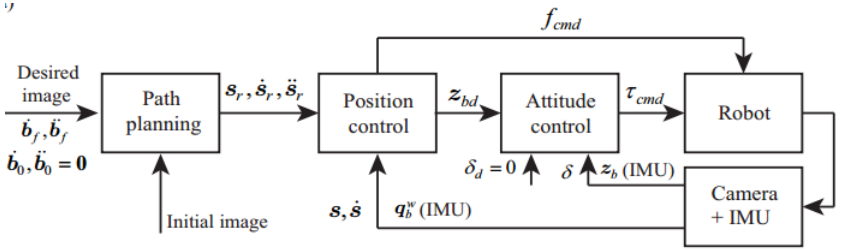
Step 3: aggressive maneuvers in eye-in-hand configuration

Plan and control the motion of aerial robot for aggressive maneuvers using an onboard camera

Outline:



Aggressively flying through a window



Proposed control diagram [J3]

- Trajectory planning generates dynamically feasible trajectory in the image coordinate based on differential flatness and homography theory
- Image-based control scheme consists of an image-based position-control loop with a quaternion-based attitude-control loop.

Related publications:

[J3] **D. Guo** and K. K. Leang, " Image-based Estimation, Planning, and Control for High-speed Flying through Multiple Openings ", *The International Journal of Robotics Research*, 2019. **Under review.**

Step 3: aggressive maneuvers in eye-in-hand configuration

Design the motion planning strategy and control scheme

- **Dynamics:**

Image Jacobian

Passivity-like image feature

Linear velocity

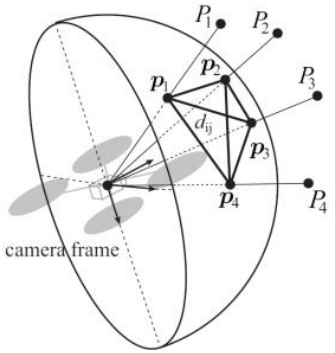
$$\dot{r} = \mathbf{J}_r \boldsymbol{V}$$

Translational $m\dot{\boldsymbol{V}} = mg\boldsymbol{E}_3 - \boldsymbol{q}_b^w \odot \boldsymbol{E}_3 f$ $\Rightarrow$ $m\mathbf{J}_r^+ \ddot{r} = -m\dot{\mathbf{J}}_r^+ \dot{r} + mg\boldsymbol{E}_3 - \boldsymbol{q}_b^w \odot \boldsymbol{E}_3 f$

Angular $\dot{\boldsymbol{q}}_b^w = 1/2 \boldsymbol{q}_b^w \otimes \boldsymbol{q}(\boldsymbol{\omega})$

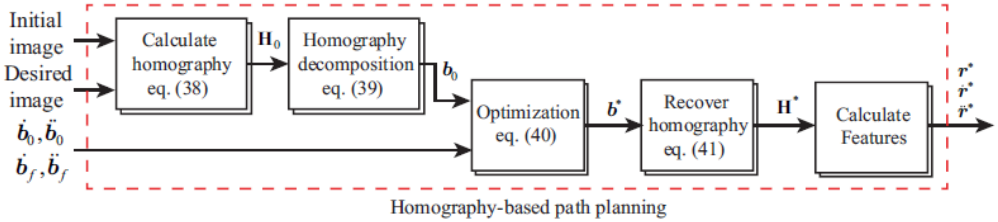
$\mathcal{I}\dot{\boldsymbol{\omega}} = -sk(\boldsymbol{\omega})\mathcal{I}\boldsymbol{\omega} + \boldsymbol{\tau}$

It can be shown that the system is differential flatness and $\boldsymbol{r}$ is the flat outputs

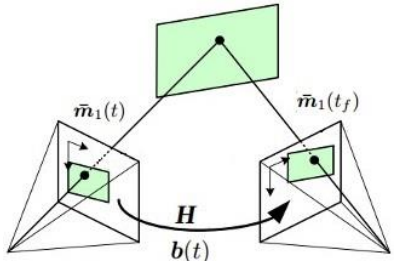


Unified sphere coordinate

- **Trajectory generation:**



Homography-based path planning



Homography matrix $\mathbf{H}$

Minimize snap of translation

$$\min \int_0^{t_f} \boldsymbol{b}^{(4)T}(t)\boldsymbol{b}^{(4)}(t)dt$$

$$\text{s.t. } \mathbf{H}(0) = \mathbf{H}_0, \mathbf{H}(t_f) = \mathbf{I}$$

$$\boldsymbol{b}^{(\kappa)}(0) = \boldsymbol{b}_0^{(\kappa)}, \boldsymbol{b}^{(\kappa)}(t_f) = \boldsymbol{b}_f^{(\kappa)}, \kappa = 1, \ldots, 4$$

Differentially flat property

$\boldsymbol{b}^*(t)$

Optimal homography trajectory

$$\mathbf{H}^*(t) = \mathbf{R}_0 e^{[\boldsymbol{\theta}_0]t/t_f} + \frac{\boldsymbol{b}^*(t)}{d_f}\boldsymbol{\iota}_f$$

Optimal trajectory of image feature

$\boldsymbol{r}^*$

Step 3: aggressive maneuvers in eye-in-hand configuration

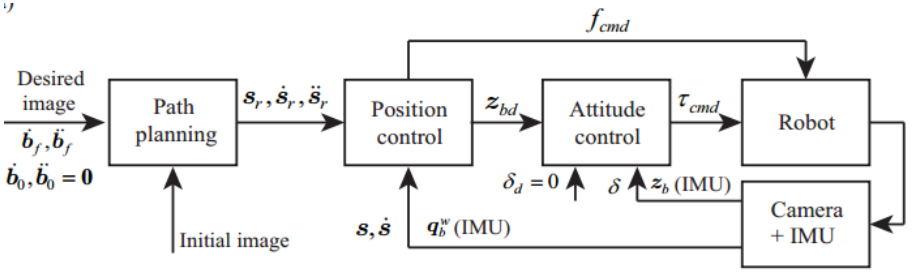
- **Image-based position controller:**

Image errors: $\boldsymbol{e}_r = \boldsymbol{r} - \boldsymbol{r}^*$

Commanded thrust: $f_{cmd} = -\boldsymbol{f}_d \cdot \boldsymbol{z}_b$ (z-axis of body frame)

$$\boldsymbol{f}_d = -m\mathbf{J}_r^{+}(\underbrace{\mathbf{K}_p\boldsymbol{e}_r + \mathbf{K}_d\dot{\boldsymbol{e}}_r + \mathbf{K}_I\int \boldsymbol{e}_r dt}_{\text{PID feedback}}) + \underbrace{m(\dot{\mathbf{J}}_r^{+}\dot{\boldsymbol{r}} + \mathbf{J}_r^{+}\ddot{\boldsymbol{r}}^* - g\boldsymbol{E}_3)}_{\text{Feedforward}}$$

The control loop is globally asymptotically stable



- **Quaternion-based attitude controller:**

Error quaternion: $\boldsymbol{q}_{e1} = \begin{bmatrix} \cos(\alpha/2) \\ {}^b\boldsymbol{n}\sin(\alpha/2) \end{bmatrix}$ where

Difference between current and desired z-axis

$$\alpha = \arccos(\boldsymbol{z}_b \cdot \boldsymbol{z}_{bd})$$

$${}^b\boldsymbol{n} = \boldsymbol{q}_w^b \odot {}^w\boldsymbol{n} = \boldsymbol{q}_w^b \odot \frac{\boldsymbol{z}_b \times \boldsymbol{z}_{bd}}{||\boldsymbol{z}_b \times \boldsymbol{z}_{bd}||}$$

Rotation axis

Desired z-axis

$$\boldsymbol{z}_{bd} = -\frac{\boldsymbol{f}_d}{||\boldsymbol{f}_d||}$$

Desired angular velocity:

$$\begin{bmatrix} \omega_{bd,x} \\ \omega_{bd,y} \end{bmatrix} = 2k_{xy}\,\mathrm{sgn}(q_{e1,w}) \begin{bmatrix} q_{e1,x} \\ q_{e1,y} \end{bmatrix}$$, and the rotation along $\boldsymbol{z}_b$ is controlled independently

Commanded torque:

$$\boldsymbol{\tau}_{cmd} = \underbrace{-\mathbf{K}_\tau\mathcal{I}(\boldsymbol{\omega} - \boldsymbol{\omega}_{bd})}_{\text{Feedback}} + \underbrace{sk(\boldsymbol{\omega})\mathcal{I}\boldsymbol{\omega}}_{\text{Feedforward}}$$

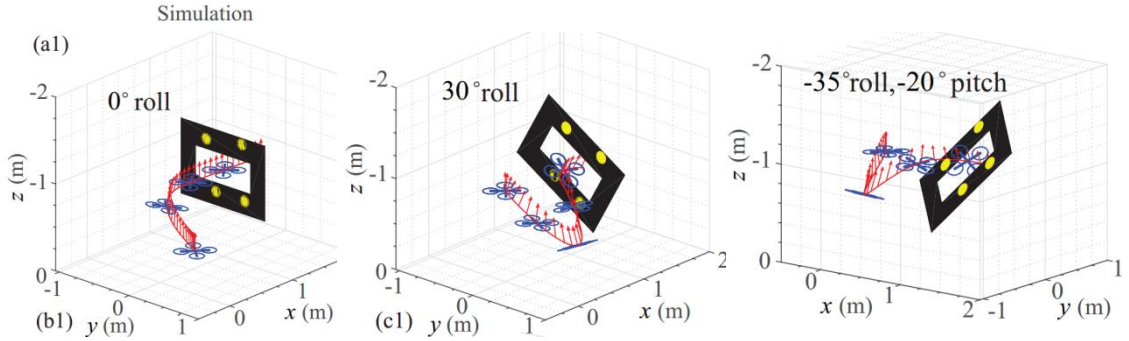
The control loop is globally asymptotically stable

Step 3: aggressive maneuvers in eye-in-hand configuration

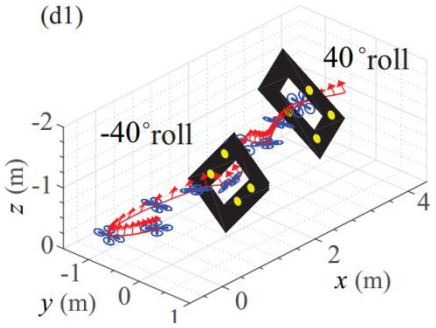
Validate performance through simulations and physical experiments

Simulations were conducted in MATLAB software.

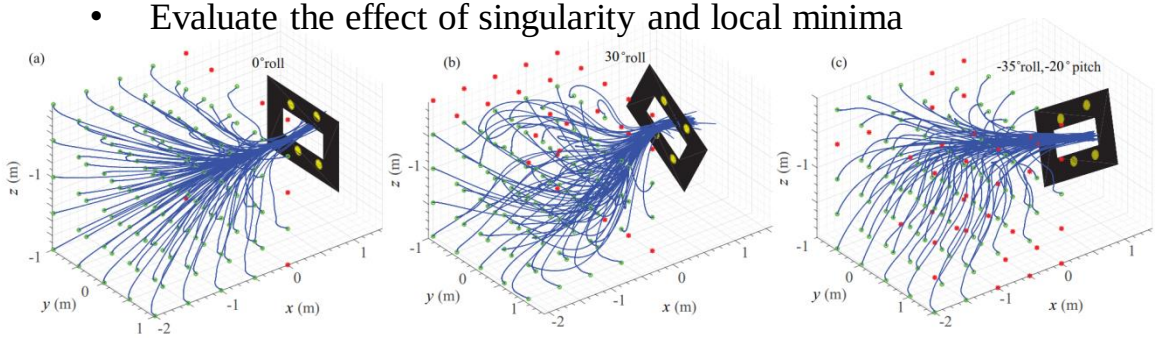
- Simulate flights through different orientations of openings

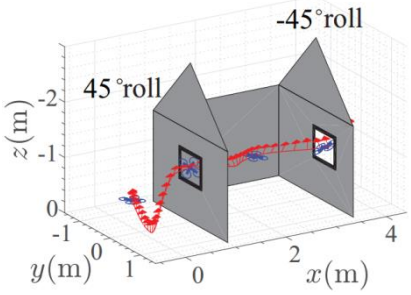


- Simulate flights through two openings



- Evaluate the effect of singularity and local minima

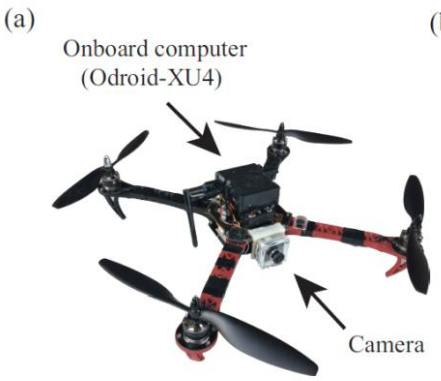




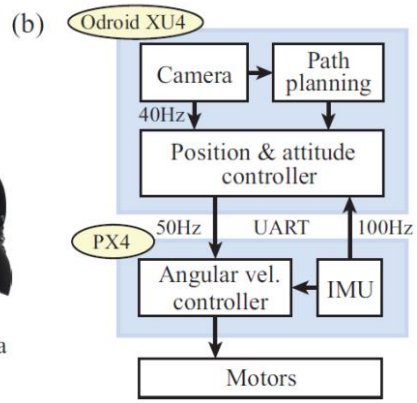
Step 3: aggressive maneuvers in eye-in-hand configuration

Experiments

- Develop the experimental platform



Experimental platform



System architecture

Table 1. Tabulated errors for simulations and experiments.

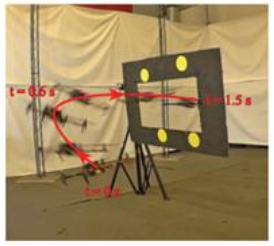
		Averaged image errors $\bar{e}_s$							
Simulation	$\bar{e}_{max}$	0°roll	0.02	30°roll	0.03	-35°roll,-20 pitch	0.04	Two windows	0.04
	$\bar{e}_{rms}$ (%)		8.18		9.73		11.14		11.46
Experiment	$\bar{e}_{max}$		0.10		0.12		0.13		0.13
	$\bar{e}_{rms}$ (%)		15.26		20.12		22.33		22.64

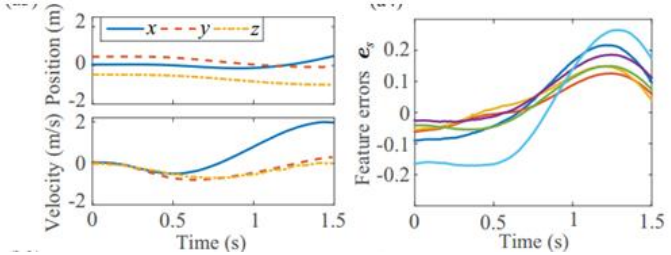
Related publication [J3] **D. Guo** and K. K. Leang, " Image-based Estimation, Planning, and Control for High-speed Flying through Multiple Openings ", *The International Journal of Robotics Research*, 2019. **Under review.**

Step 3: motion plan and control for aggressive maneuvers

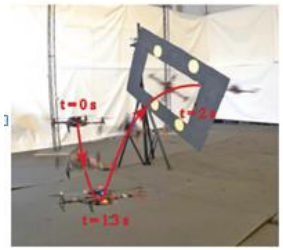
- Flying through openings with different orientations

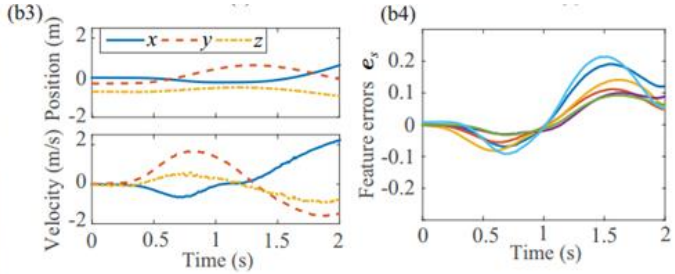
0°roll



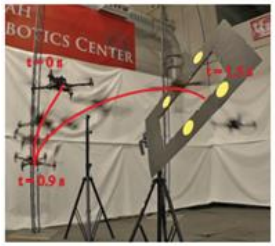


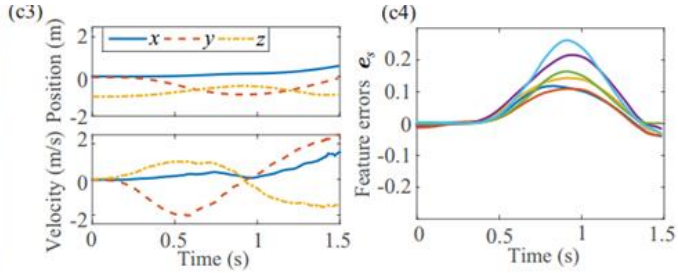
30°roll

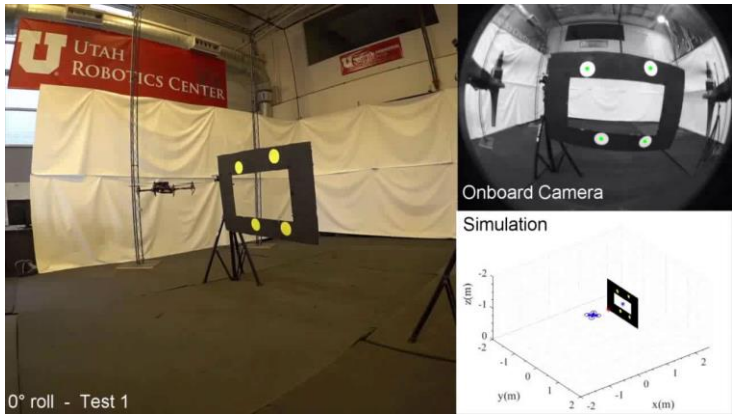


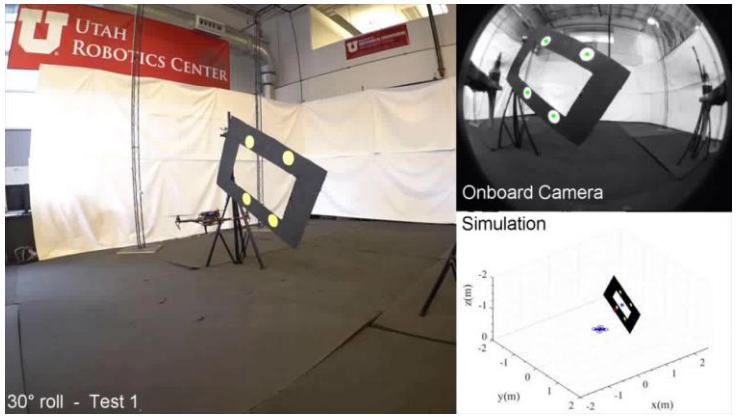


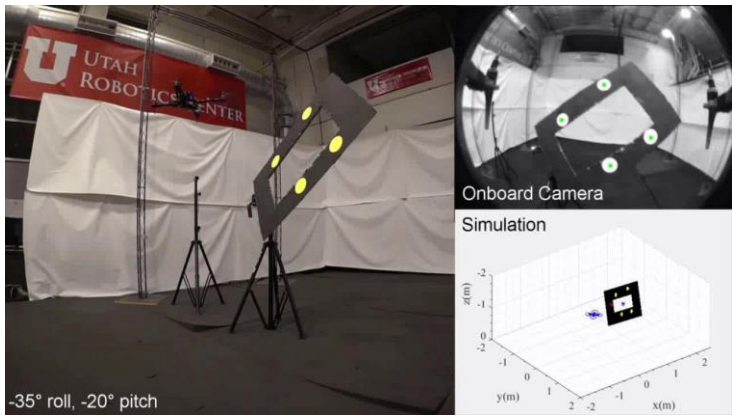
-35°roll, -20°pitch







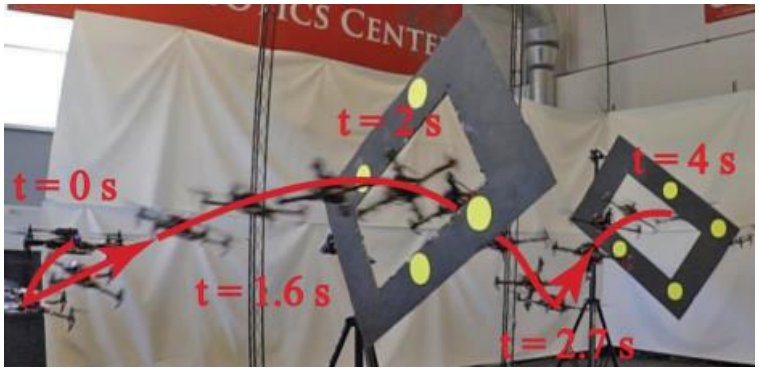


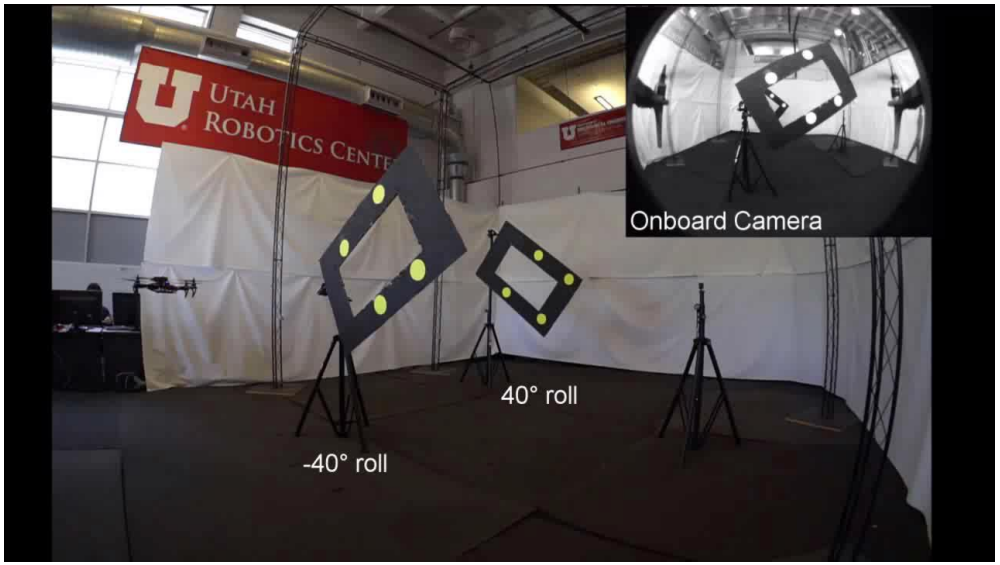


Step 3: aggressive maneuvers in eye-in-hand configuration

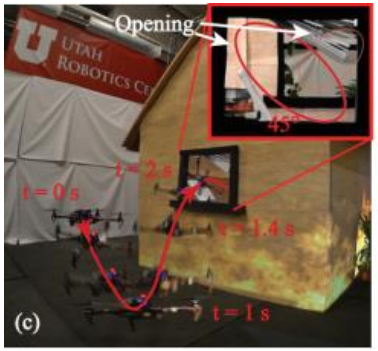
- Flights through two openings

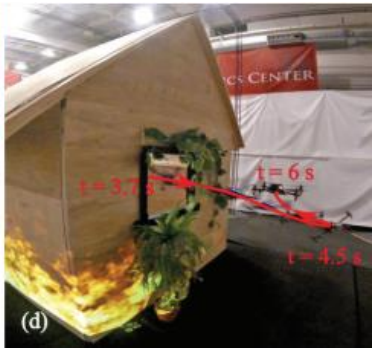
Two gaps

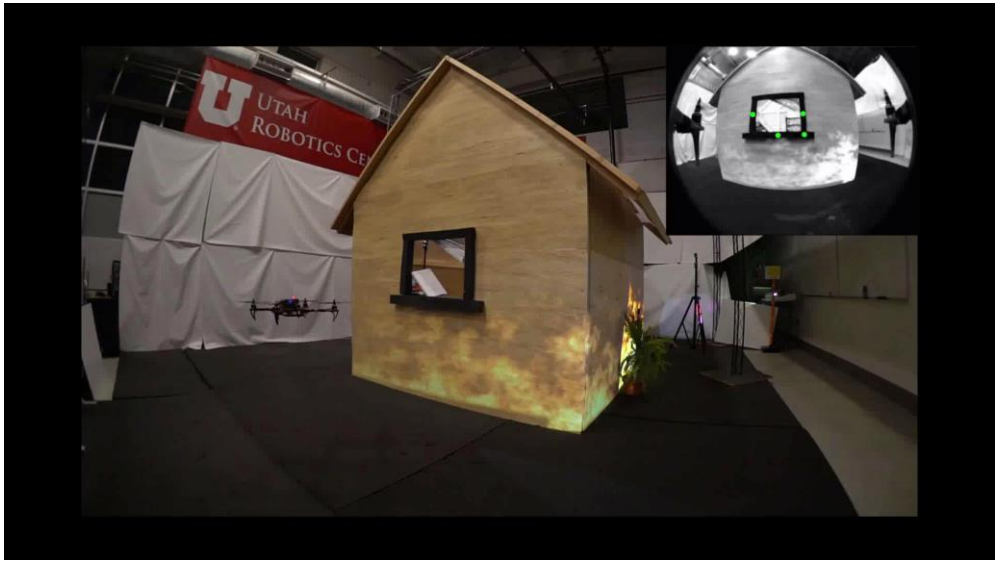




house







Conclusion: Aggressive flying through multiple openings can be achieved, where computation is performed in real-time. First demonstration of image-based aggressive maneuvering.

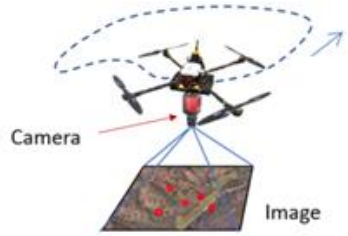
Overview of my work

What I've contributed: (1) developed image-based algorithms for estimation, planning, and control of aerial robots using a single camera; (2) provided rigorous stability and performance analysis; and (3) validated through experiments

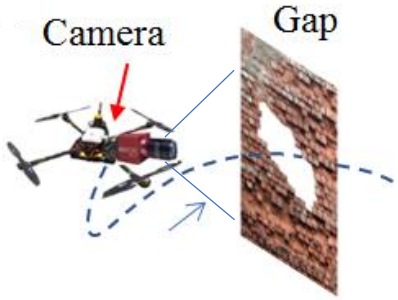


Mobile camera √

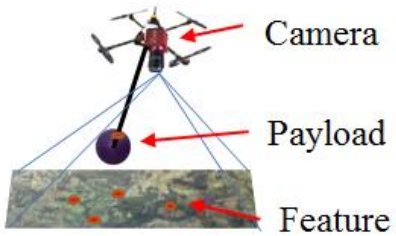
Step1: estimate and control through a mobile overhead camera (Eye-to-hand configuration + IBVS)



Step2: localize and control through an onboard camera (Eye-in-hand configuration + PBVS)



Step3: motion plan and control for aggressive maneuvering through openings (Eye-in-hand configuration + IBVS)



Step4: motion plan and control to carry payloads (Eye-in-hand configuration + IBVS)



Step 4: carrying payload in eye-in-hand configuration

Prior work and their limitations:

- Motion capture system → Flights with a cable-suspended payload

  - **Methods**: Motion planning using differential flatness [Sreenath et. al., 2013]/ reinforcement learning [Faust et. al., 2017]; Nonlinear hierarchical control [Liang et. al., 2018]
  - **Limitations**: ▪ Rely on motion capture systems
    ▪ Require translational variables of both the robot and payload



[Liang et. al., 2018]



[Sreenath et. al., 2013]

- Onboard camera → Flights with a cable-suspended payload
  - **Methods**: Same control approach + pose estimation of payload using the camera [Tang et. al., 2018]
  - **Limitations**: ▪ Rely on motion capture systems
    ▪ Require translational variables of both the robot and payload



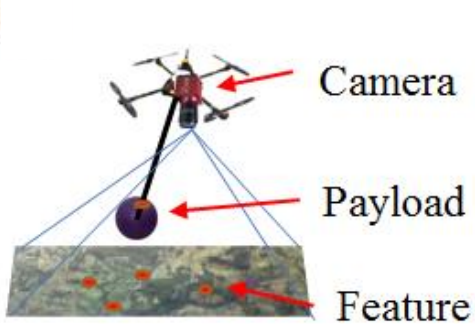
[Tang et. al., 2018]

Knowledge gap: Limited work have resulted flying with a cable-suspended payload using only an onboard camera and without considering translational variables.

Step 4: carrying payload in eye-in-hand configuration

My approach: Image-based motion planning and control of aerial robots for transport a cable-suspended payload. (Eye-in-hand configuration + IBVS)

My results:



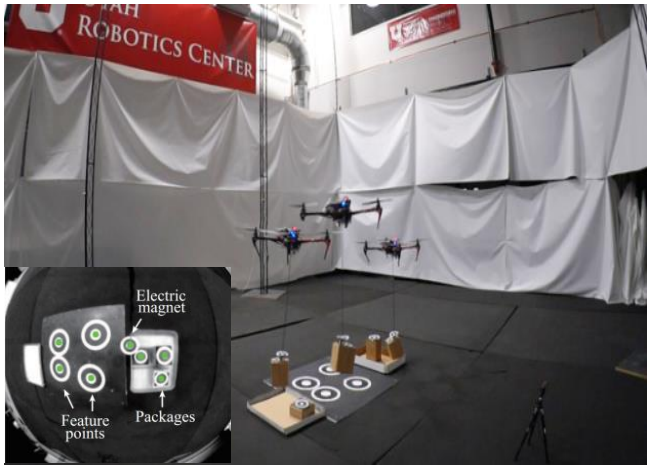


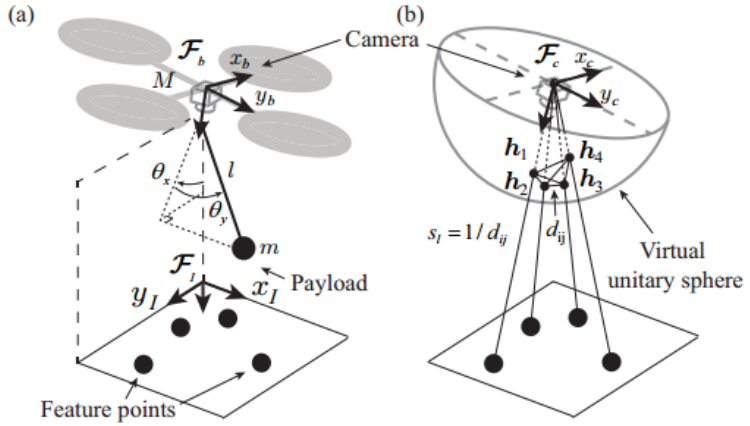
Image from onboard camera

Contribution: First demonstration of motion planning and control in the image coordinate of an onboard camera to transport cable-suspended payloads independent of external positioning systems.

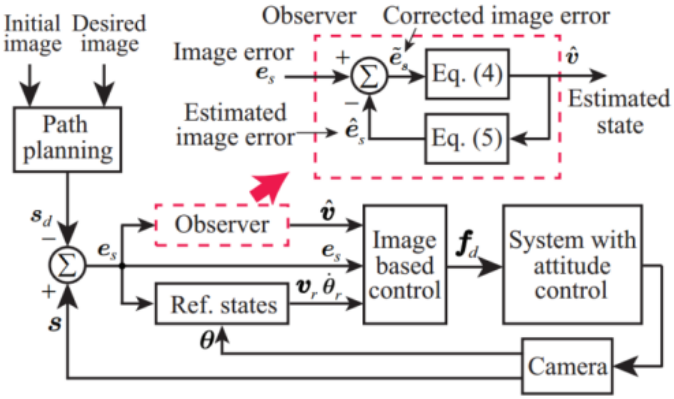
Step 4: carrying payload in eye-in-hand configuration

Plan the motion and control an aerial robot for flying with a cable-suspended payload using an onboard camera

Outline:



Flying with a suspended payload



Proposed control diagram [J4]

- Trajectory planning generates dynamically feasible trajectory in the image coordinate based on differential flatness and homography theory
- Image-based control scheme consists of an image-based position-control loop with a quaternion-based attitude-control loop.

Related publications:

[J4] **D. Guo** and K. K. Leang, "Image-Based Estimation, Planning, and Control of a Cable-Suspended Payload for Package Delivery", *IEEE Robotics and Automation Letters*, 2020, Accepted.

Step 4: carrying payload in eye-in-hand configuration

Extend the method to the robot-payload case and validate algorithms through demonstrations

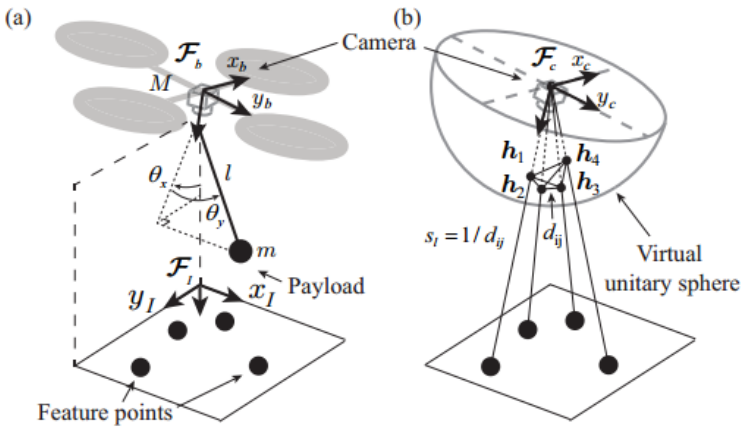
- **Objective:** Estimate linear velocity; Reduce image and velocity tracking error; Minimize swing angles

- **Dynamics:** (States, Thrust, Position, Swing angles)

Translational: $\mathbf{H}(\boldsymbol{\theta})\ddot{\boldsymbol{q}} + (\mathbf{C}(\boldsymbol{\theta},\dot{\boldsymbol{\theta}}) + \mathbf{D})\dot{\boldsymbol{q}} + \mathbf{G}(\boldsymbol{\theta}) = \boldsymbol{u}, \qquad \boldsymbol{q} = [\boldsymbol{p}^T, \boldsymbol{\theta}^T]^T$

Angular: $\mathcal{I}\dot{\boldsymbol{\Omega}}_b + sk(\boldsymbol{\Omega}_b)\mathcal{I}\boldsymbol{\Omega}_b = \boldsymbol{\tau}.$

Image feature: $\dot{\boldsymbol{s}} = \mathbf{J}\boldsymbol{v}$ (Image Jacobian) Image errors: $\boldsymbol{e_s} = \boldsymbol{s} - \boldsymbol{s}_d$



- **Observer:** (Corrected image error)

Estimated velocity: $\hat{\boldsymbol{v}} = \boldsymbol{w} + \mathbf{K}_2\mathbf{J}^{+}\tilde{\boldsymbol{e}}_{\boldsymbol{s}}, \qquad (4)$

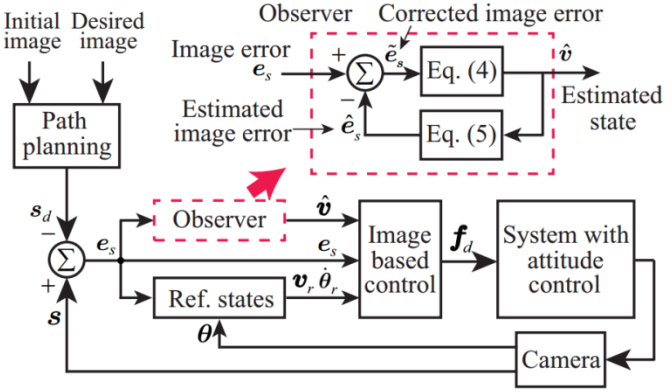
Estimated image error: $\dot{\hat{\boldsymbol{e}}}_{\boldsymbol{s}} = \mathbf{J}\hat{\boldsymbol{v}} + \mathbf{K}_1\tilde{\boldsymbol{e}}_{\boldsymbol{s}} - \dot{\boldsymbol{s}}_d, \qquad (5)$

$\dot{\boldsymbol{w}} = \boldsymbol{a} + \mathbf{K}_2\mathbf{B}\tilde{\boldsymbol{e}}_{\boldsymbol{s}} + \mathbf{J}^T\tilde{\boldsymbol{e}}_{\boldsymbol{s}} - \mathbf{K}_v\hat{\boldsymbol{e}}_v - \mathbf{K}_p\boldsymbol{\delta}_v$

- **Position controller:**

Referenced velocity: $\boldsymbol{v}_r = \mathbf{J}^{+}(-\lambda\boldsymbol{e_s} + \dot{\boldsymbol{s}}_d)$

Force: $\boldsymbol{f}_d = -\underbrace{\mathbf{J}^T\mathbf{K}_s\boldsymbol{\sigma}_s - \mathbf{K}_p\boldsymbol{\delta}_v}_{\text{PD}} - \underbrace{\mathbf{K}_v\hat{\boldsymbol{e}}_v}_{\text{I}} + \mathbf{H}_1\ddot{\boldsymbol{q}}_r + (\mathbf{C}_1 + \mathbf{D}_1)\dot{\boldsymbol{q}}_r + \mathbf{G}_1$



- **Stability analysis:** the system is asymptotically stable

$V = V_1 + V_2 \qquad V_1 = \frac{1}{2}(\tilde{\boldsymbol{e}}_{\boldsymbol{v}}^T\tilde{\boldsymbol{e}}_{\boldsymbol{v}} + \tilde{\boldsymbol{e}}_{\boldsymbol{s}}^T\tilde{\boldsymbol{e}}_{\boldsymbol{s}}), \; V_2 = \frac{1}{2}(\boldsymbol{e}_{\dot{\boldsymbol{q}}}^T\mathbf{H}\boldsymbol{e}_{\dot{\boldsymbol{q}}} + \boldsymbol{\delta}_{\boldsymbol{v}}{}^T\mathbf{K}_{\boldsymbol{p}}\boldsymbol{\delta}_{\boldsymbol{v}})$

$\Rightarrow \dot{V} \leq 0$ Barbalats Lemma $\Rightarrow \boldsymbol{e_s}, \dot{\boldsymbol{e}}_{\boldsymbol{s}}, \boldsymbol{e}_{\boldsymbol{v}} \to \boldsymbol{0}$ as $t \to \infty$

Related publication [J4] **D. Guo** and K. K. Leang, "Image-Based Estimation, Planning, and Control of a Cable-Suspended Payload for Package Delivery", *IEEE Robotics and Automation Letters*, 2020, Accepted.

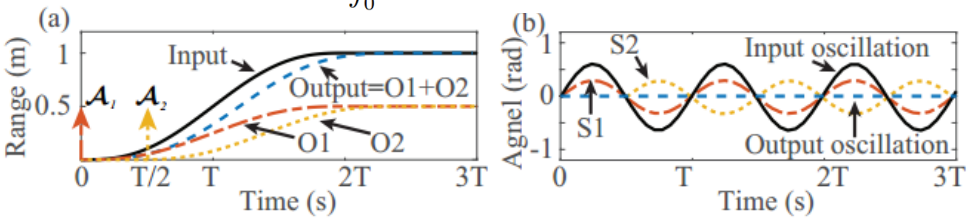
Step 4: carrying payload in eye-in-hand configuration

- **Trajectory planning:**

  Differential flatness: The set $\mathcal{Y} = \{\boldsymbol{s}, \boldsymbol{\theta}, \psi\}$ is the flat output $\boldsymbol{\theta}_d$=0

  Homography trajectory optimization: $\min \int_0^{t_f} \boldsymbol{b}^{(4)T}(t)\boldsymbol{b}^{(4)}(t)dt$

  Sway reduction (ZV input shaping): $T = 2\sqrt{l/g}$

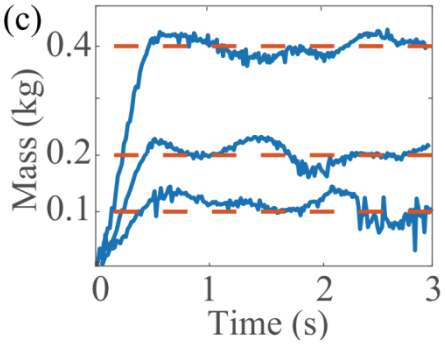


- **Adapt unknown payload's mass:**

  Least-square identification:

  Simplified dynamics: $-(M+m)g = \boldsymbol{f} \cdot \boldsymbol{z}_I$

  Update law: $\hat{m} = -\frac{1}{gc}\sum_{i=1}^{c} \boldsymbol{f}(i) \cdot \boldsymbol{z}_I - M$

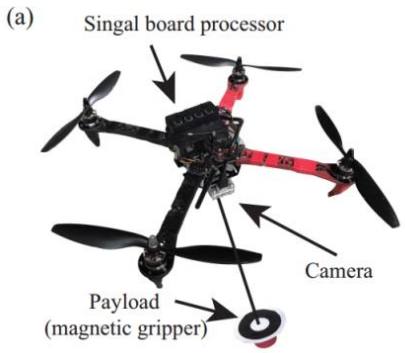


Related publication [J4] **D. Guo** and K. K. Leang, "Image-Based Estimation, Planning, and Control of a Cable-Suspended Payload for Package Delivery", *IEEE Robotics and Automation Letters*, 2020, Accepted.

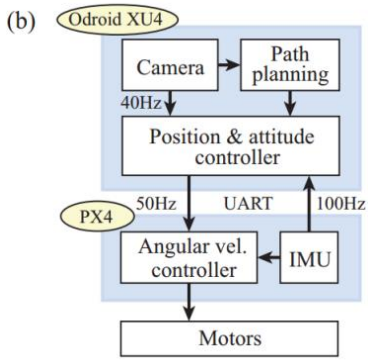
Step 4: carrying payload in eye-in-hand configuration

Validate performance through physical experiments

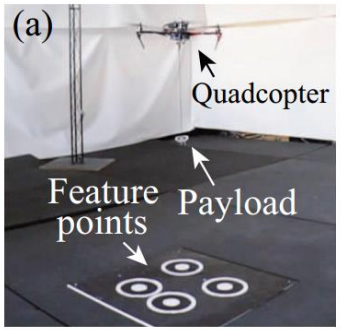
- Develop the experimental platform

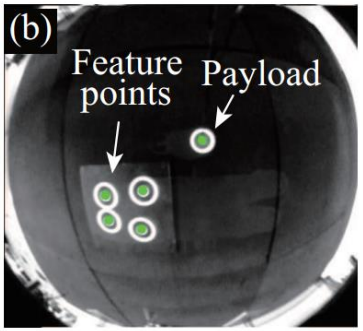


Experimental platform



System architecture



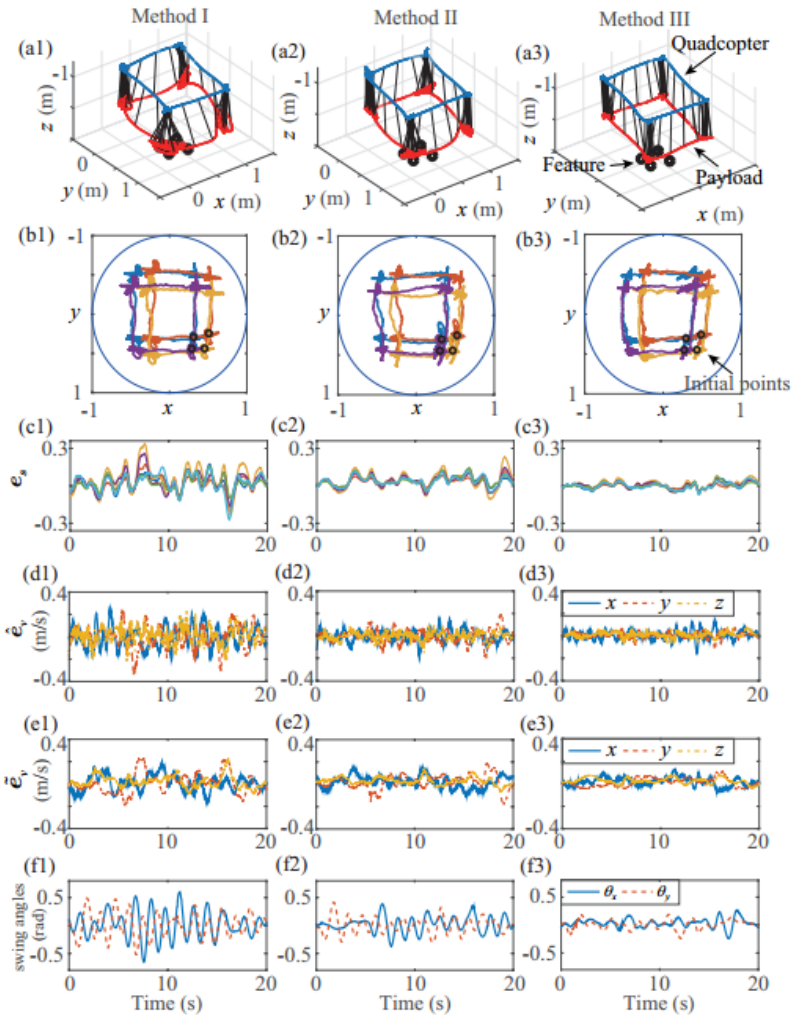


- Square trajectory

Method I: Image-based control without feedback of swing angles
Method II: Image-based control with feedback of swing angles
Method III: Method II + input shaper

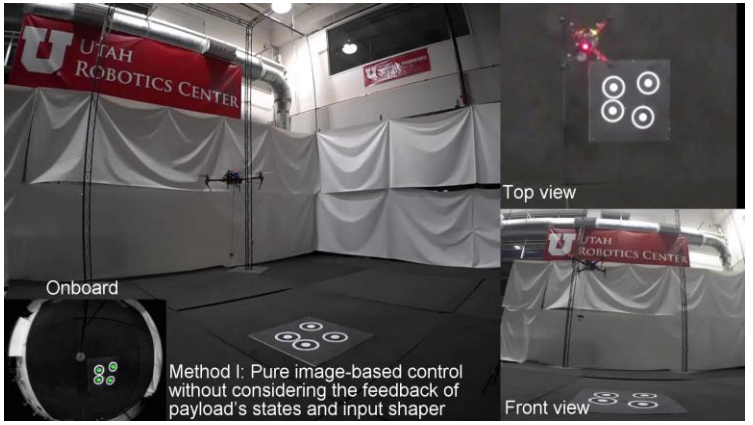
Related publication [J4] **D. Guo** and K. K. Leang, "Image-Based Estimation, Planning, and Control of a Cable-Suspended Payload for Package Delivery", *IEEE Robotics and Automation Letters*, 2020, Accepted.

Step 4: carrying payload in eye-in-hand configuration

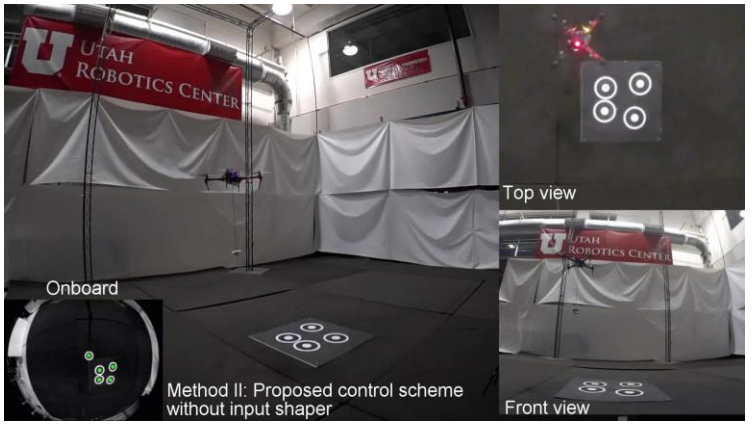


	Square trajectory			
	e_s		θ	
	max	rms (%)	max (rad)	rms (%)
I	0.32	34.28	0.66	39.32
II	0.21	28.58	0.46	29.33
III	0.08	22.86	0.27	12.47

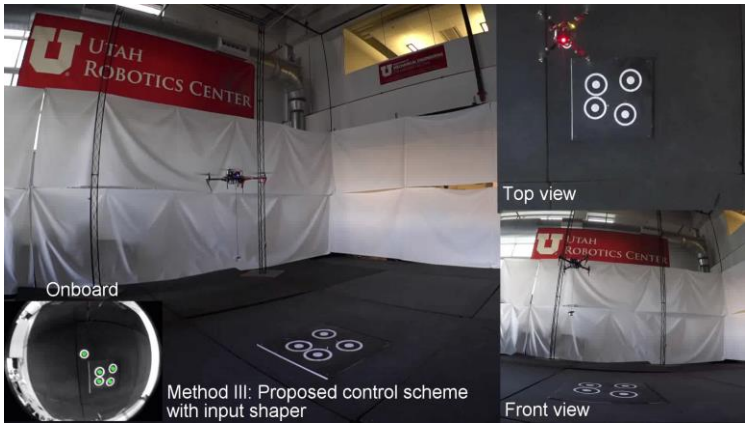
Method I



Method II

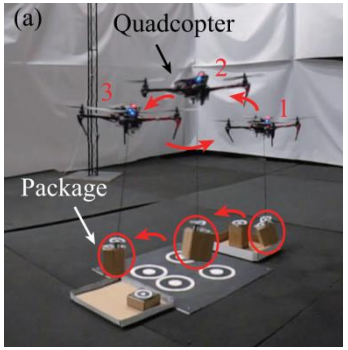


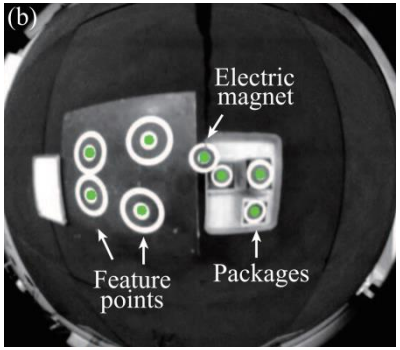
Method III



Step 4: carrying payload in eye-in-hand configuration

- Package delivery



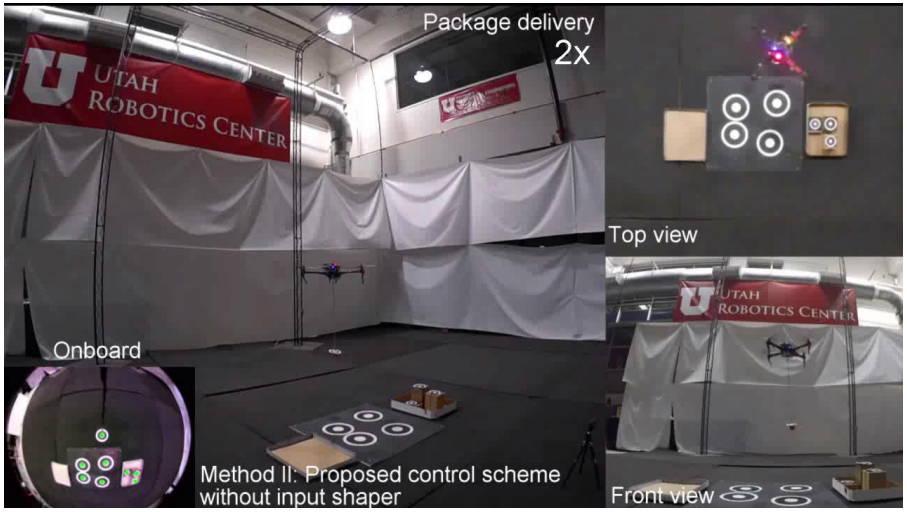


	Package delivery			
	e_s		θ	
	max	rms (%)	max (rad)	rms (%)
I	N/A	N/A	N/A	N/A
II	0.22	31.43	0.50	34.77
III	0.14	24.59	0.24	16.98

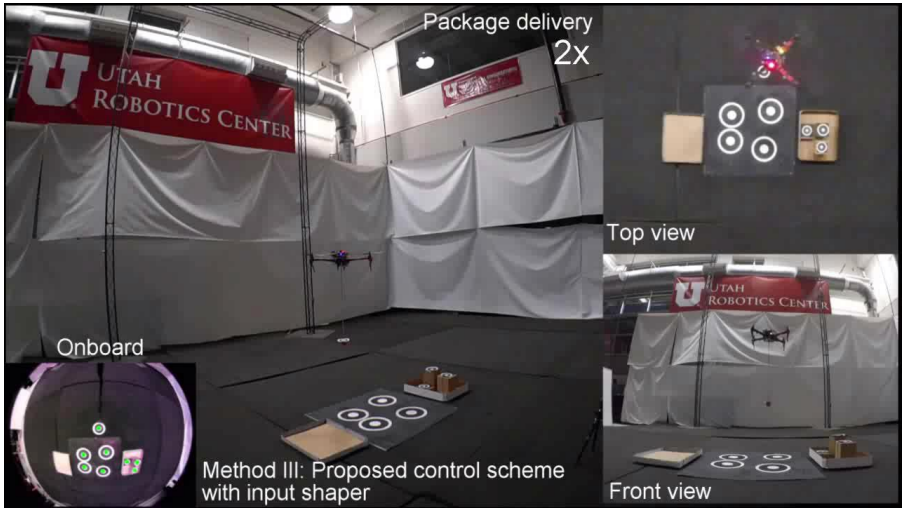
Related publication [J4] **D. Guo** and K. K. Leang, "Image-Based Estimation, Planning, and Control of a Cable-Suspended Payload for Package Delivery", *IEEE Robotics and Automation Letters*, 2020, Accepted.

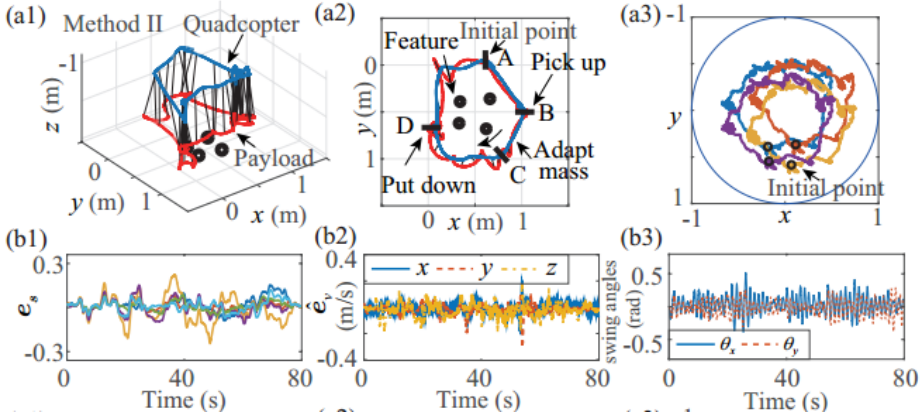
Step 4: carrying payload in eye-in-hand configuration

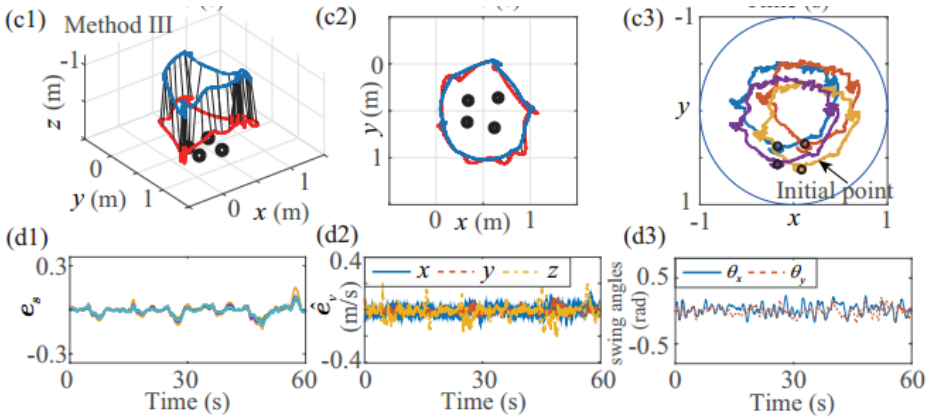
Method II



Method III







Conclusion: The image-based trajectory planning and control algorithms enable aerial robots to pick up and transport payloads with unknown mass by minimizing the swing angles.

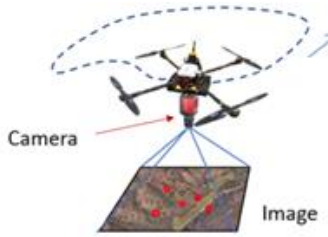
Conclusions

Localization and control through mobile overhead camera is possible, but sensitive to camera's trajectory



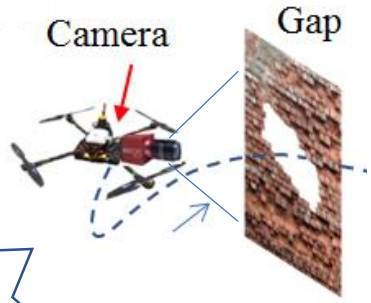
Mobile camera √

Step1



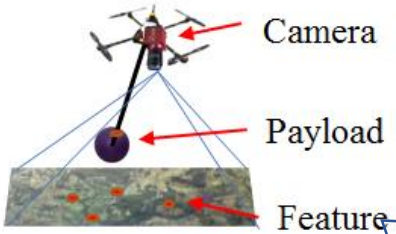
Step2

Localization and control through onboard camera is possible; however only validated in slow-motion



Step3

Image-based high-speed maneuvers through openings is possible, but limited to FOV and blurred image



Step4

Image-based payload delivery is possible; but only validated in slow-motion

Future work

Future work:

- **Overhead camera**: Develop dynamic control using more than one camera which move in general trajectories; develop trajectory planning method to generate smooth trajectory with considering field of view (FOV).

- **Onboard camera**: Develop new robust passivity-like image features; consider FOV in trajectory planning; avoid using desired images; consider moving referenced targets; cooperative aerial manipulation by multiple robots

Related publications

[J1] **D. Guo**, J. Bourne, H. Wang, W. Yim, and K. K. Leang, "Adaptive-repetitive visual-servo control of low-flying aerial robots via uncalibrated high-flying cameras", *Journal of Nonlinear Science, Special Issue on Robotics: Mechanics and Control of Locomotion*, 27(4): 1235 – 1256, 2017.
[J2] **D. Guo**, H. Wang, and K. K. Leang. "Nonlinear vision-based observer for visual servo control of an aerial robot in GPS-denied environments", *ASME Journal of Mechanisms and Robotics*, 10(6), 061017-061017-13, 2018.

[J3] **D. Guo** and K. K. Leang, "Image-based Estimation, Planning, and Control for High-speed Flying through Multiple Openings ", *The International Journal of Robotics Research*, 2019. **Under review.**

[J4] **D. Guo** and K. K. Leang, “Image-Based Estimation, Planning, and Control of a Cable-Suspended Payload for Package Delivery”, *IEEE Robotics and Automation Letters*, 2020, Accepted.

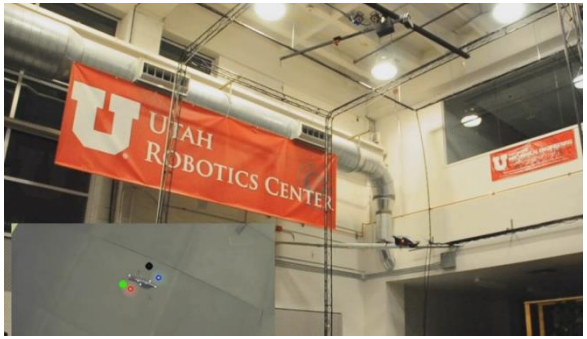
[C1] **D. Guo**, W. Yim, and K. K. Leang, "Adaptive repetitive visual-servo control of a low-flying unmanned aerial vehicle with an uncalibrated high-flying camera", *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Daejeon, Korea, Oct. 9-13, 2016.

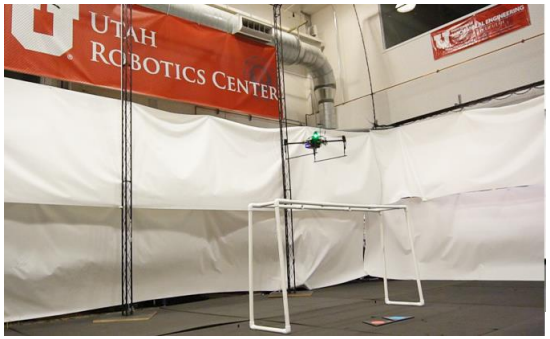
[C2] X. He, **D. Guo**, and K. K. Leang, "Repetitive control design and implementation for periodic motion tracking in aerial vehicle", *American Control Conference (ACC)*, Seattle, WA, USA, May 24-26, 2017.

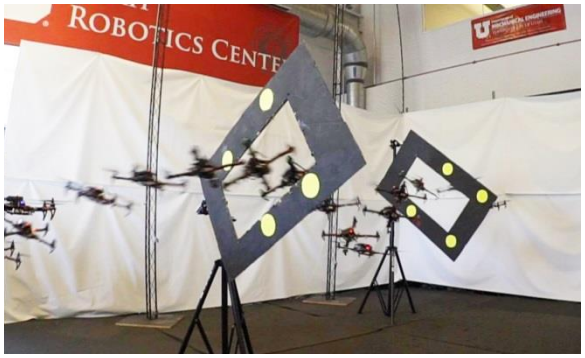
[C3] **D. Guo** and K. K. Leang. "Position and linear velocity estimation for position-based visual servo control of an aerial robot in GPS-denied environments", *ASME Dynamic Systems and Control Conference (DSCC)*, Tysons Corner, Virginia, USA, Oct. 11-13, 2017. (**This paper won the 2017 ASME DSCC Student Best Paper Award**)

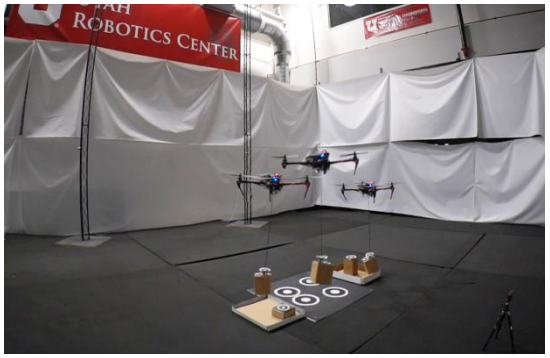
THE UNIVERSITY OF UTAH DARCLab DESIGN, AUTOMATION, ROBOTICS & CONTROL

ROBOTICS CENTER THE UNIVERSITY OF UTAH









Thank you!